MIL-STD -1598
7 April 1978

# MILITARY STANDARD

# STUDS PREFERRED FOR DESIGN, LISTING OF

FSC 5307

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DEPARTMENT OF DEFENSE
Washington, DC 20301

Studs Preferred For Design, Listing of:

MIL-STD-1598

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Aeronautical Systems Division (AFSC), ATTN: ASD/ENESS, Wright-Patterson Air Force Base, Ohio 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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# FOREWORD

1. The purpose of this bookform standard is to provide a commodity type parts document on STUDS to aid military equipment designers and engineers in the selection of preferred studs.

2. This document consists of an index of preferred standardization documents and a listing of preferred parts within these documents that have been selected with respect to configuration, sizes, lengths, materials, and finishes for studs.

3. The selection of preferred documents listed in this standard and the selection of part numbers within the preferred documents were made as follows:

a. Selection of Documents

(1) Documents suitable for listing in the Department of Defense Index of Specifications and Standards (DODISS).

(2) Documents which are active for design.

(3) Documents specifying part numbers (dash numbers) which designate specific sizes, materials and finishes.

b. Selection of Part Numbers

(1) By conducting a thorough search and evaluation of existing DoD procurement information.

(2) By evaluation of preferred parts listed in recent weapon system contracts.

(3) By evaluation of preferred parts lists obtained from industry.

4. To increase the scope and versatility of this stud standard, periodic revisions will be developed. Results from Standardization studies, MILITARY PARTS CONTROL ADVISORY GROUP (MPCAG) evaluations, evaluation of a new family of studs and recommendations from interested activities will form the basis for these revisions.



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# 1. SCOPE

1.1 Scope. This standard provides a listing of preferred studs encompassing the following characteristics:

a. Configuration

b. Size

c. Materials

d. Protective Coatings and Finishes

1.2 Purpose. The purpose of this standard is as follows:

a. provide the designer with a listing of preferred studs to promote their use in design of weapon systems and equipments.

b. control and minimize the variety of studs used in military equipment thereby facilitating logistic support of the equipment during its life cycle.

1.3 Application. To minimize the proliferation of studs, only the preferred part numbers listed herein are authorized for use in new design. All other part numbers, even though shown on current Military Specification Sheets, Military Standards (MS), National Aerospace Standards (NAS), Aeronautical Standards (AS), and Air Force/Navy Aeronautical Standards (AN), are not approved for use in new design.

1.4 Intended use. Implement this standard by including one of the following options in the contract:

a. When an end use type standard such as MIL-STD-1515 is included in the contract, specify this standard as a supplement to it. When thus specified, only the stud part numbers listed in the end use type standard and listed herein are acceptable. Use of other studs requires approval of the Government procuring activity.

b. When an end use type standard is not included in the contract, specify this standard and list in the contract only the studs or groups of studs listed herein that are acceptable. Use of other studs requires approval of the Government procuring activity.

c. When an end use type standard is not included in the contract, specify this standard as a guide for the selection of studs.



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## 2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-45933/1 - Stud, Keyring, Locked, 125 KSI FTU, Single Step

MIL-S-45933/2 - Stud, Keyring, Locked, 160 KSI FTU, Double Step

STANDARDS

MILITARY

MS9827 - Stud - Stepped, 2 Dia. Engagement, Steel, .250-20UNJS X .190-32UNJF

MS9828 - Stud - Stepped, 2 Dia. Engagement, Steel, .3125-18UNJS X .250-28UNJF

MS9829 - Stud - Stepped, 2 Dia. Engagement, Steel, .375-16UNJS X .3125-24UNJF

MS9830 - Stud - Stepped, 2 Dia. Engagement, Steel, .4375-14UNJS X .375-24UNJF

MS9831 - Stud - Stepped, 2 Dia. Engagement, Steel, .500-13UNJS X .4375-20UNJF

MS9832 - Stud - Stepped, 2 Dia. Engagement, Steel, .5625-12UNJS X .500-20UNJF

MS9833 - Stud - Stepped, 2 Dia. Engagement, Steel, .625-11UNJS X .5625-18UNJF

MS9834 - Stud - Stepped, Drilled, 2 Dia. Engagement, Steel, .250-20UNJS X .190-32UNJF



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STANDARDS

MILITARY - Continued

MS9835 - Stud - Stepped, Drilled, 2 Dia. Engagement, Steel, .3125-18UNJS X .250-28UNJF

MS9836 - Stud - Stepped, Drilled, 2 Dia. Engagement, Steel, .375-16UNJS X .3125-24UNJF

MS9837 - Stud - Stepped, Drilled, 2 Dia. Engagement, Steel, .4375-14UNJS X .375-24UNJF

MS9838 - Stud - Stepped, Drilled, 2 Dia. Engagement, Steel, .500-13UNJS X .4375-20UNJF

MS9839 - Stud - Stepped, Drilled, 2 Dia. Engagement, Steel, .5625-12UNJS X .500-20UNJF

MS9840 - Stud - Stepped, Drilled, 2 Dia. Engagement, Steel, .625-11UNJF X .5625-18UNJF

MS16187 - Stud - Continuous-Thread-(Bolt-Stud), B16 Alloy Steel, High Temperature Service (875° - 1000°F)

MS17293 - Stud-Stepped, 1.5 Dia. Engagement, .250-20 X .190-32 (MIL-S-8879 Thread)

MS17294 - Stud-Stepped, 1.5 Dia. Engagement, .3125-18 X .250-28 (MIL-S-8879 Thread)

MS17295 - Stud-Stepped, 1.5 Dia. Engagement, .375-16 X .3125-24 (MIL-S-8879 Thread)

MS17296 - Stud-Stepped, 1.5 Dia. Engagement, .4375-14 X .375-24 (MIL-S-8879 Thread)

MS17297 - Stud-Stepped, Drilled, 1.5 Dia. Engagement, .3125-18 X .250-28 (MIL-S-8879 Thread)

MS17298 - Stud-Stepped, Drilled, 1.5 Dia. Engagement, .375-16 X .3125-24 (MIL-S-8879 Thread)

MS17299 - Stud-Straight, .190-24 X .190-32 (MIL-S-8879 Thread)



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STANDARDS

MILITARY - Continued

MS17300 - Stud-Straight, .250-20 X .250-28 (MIL-S-8879 Thread)

MS17301 - Stud-Straight, .3125-18 X .3125-24 (MIL-S-8879 Thread)

MS17302 - Stud-Straight, Drilled, .250-20 X .250-28 (MIL-S-8879 Thread)

MS17303 - Stud-Straight, Drilled, .3125-18 X .3125-24 (MIL-S-8879 Thread)

MS27951 - Stub (Stud), Welding Turnbuckle, (General Purpose, Manual Weld Application)

MS51833 - Stud, Locked In, Key Locked, Lighweight

MS51834 - Stud, Locked In, Key Locked, Heavy Duty

MS51864 - Stud, Plain, General Purpose
MS51950 - Stud, Shouldered: Wheel Mounting

MS51951 - Stud, Shouldered and Stepped: Wheel Mounting

MS51989 - Stud, Locked In, Ring Locked, Serrated

MS51992 - Stud, Locked In, Ring Locked, Serrated, High Strength

MS52122 - Stud, Shouldered and Stepped: Wheel Mounting

AIR FORCE-NAVY AERONAUTICAL

AN126587 Thru AN126652 - Stud-Straight, .375-16 X .375-24

AN126881 Thru AN127192 - Stud-Straight, Necked, .375-16 X .375-24

AN128363 Thru AN128686 - Stud-Straight, Drilled, .250-20 X .250-28

AN128687 Thru AN128998 - Stud-Straight, Drilled, .3125-18 X .3125-24

AN129293 Thru AN129604 - Stud-Straight, Drilled, Necked, .375-16 X .375-24



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STANDARDS

AIR FORCE-NAVY AERONAUTICAL - Continued

AN152601 Thru AN152900 - Stud-Stepped, Drilled, 1.5 Dia. Engagement, .250-20 X .190-32

AN158901 Thru AN159200 - Stud-Stepped, Necked, 1.5 Dia. Engagement, .4375-14 X .375-24

AN162501 Thru AN162800 - Stud-Stepped Drilled, Necked, 2 Dia. Engagement, .4375-14 X .375-24

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS1454 - Rod, Continuous Thread

(Application for copies should be addressed to the Aerospace Industries Association of America, Inc., 1725 DeSales Street, N.W., Washington, DC 20036.)

3. DEFINITIONS

3.1 Adopted Industry Standards. Any Industry Specification or Standard which is listed in this Standard/The Department of Defense Index of Specifications and Standards (DODISS).

3.2 Commodity Type Document. A document which lists preferred parts within a Federal Supply Classification class or Item Name. This document is to be used for selecting preferred parts for a new design when the document is invoked as a contractual requirement in conjunction with a parts control requirement.



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3.3 End Use Type Document. A document that lists preferred documents and establishes parts requirements which are contractually binding for the design and construction/manufacture of a weapon system or an established equipment category such as MIL-STD-1515.

3.4 Military Parts Control Advisory Group (MPCAG). A Department of Defense organization which provides advice to the Military Departments and military contractors on the selection of parts in assigned commodity classes, and collects data on nonstandard parts for developing or updating military specifications and standards.

3.5 Definitions of approved item names used in this standard are as follows:

a. Stud, Continuous Thread. A headless fastener not exceeding 12 inches in length having an external thread of one nominal diameter running from end to end.

b. Stud, Locked In. A headless fastener externally threaded on both ends with each threaded portion of one but not necessarily the same nominal diameter. It is designed to be locked in place by either a RING, LOCK, SERRATED or a RING, LOCK, KEYED.

c. Stud, Plain. A headless fastener not exceeding 12 inches in length, threaded externally on both ends with threads of the same nominal diameter. The maximum diameter of the unthreaded portion must not be greater than the nominal thread diameter plus 0.020 inches. The unthreaded portion must be without locking or wrenching provisions. It may be thread relieved.

d. Stud, Stepped. A headless fastener not exceeding 12 inches in length, threaded externally on both ends on which the nominal diameter of one threaded section is greater than the other. The entire unthreaded portion must be of a single diameter equal to or smaller than the maximum major diameter of the larger threaded end. Includes radial grooves or thread reliefs. Excludes items with tapered threads.

e. Stud, Welding. A headed or headless fastener threaded externally on one end only and intended for fastening the unthreaded end or head by welding. If headed, head is not designed to be self-holding, nor to be turned with a wrench or inserted driver, nor does the head have small raised projections on the top or underside to facilitate welding.



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4. GENERAL STATEMENTS

4.1 Selection procedure.

4.1.1 Document selection. The applicable section shall be selected after reviewing the table of contents.

4.1.2 Part number selection (preliminary). A preliminary selection of the applicable part number shall be made after reviewing the nominal parameters (sizes, materials, tensile strength or hardness) listed in the sections.

4.1.3 Part number selection (final). A final selection of the applicable part number shall be made after reviewing the detailed requirements specified in the referenced stud documents for suitability in the particular military equipment being designed (considering the application and environmental conditions).

5. DETAILED REQUIREMENTS

5.1 The detailed requirements for preferred studs are contained in the applicable stud document and associated procurement specification. If there is disagreement between the nominal parameters listed in this standard and the parameters specified in the applicable stud document or associated procurement specification, the parameters specified in the applicable stud document or associated procurement specification shall prevail.

6. NOTES

6.1 Dimensions. Dimensions shown in the sections contained herein are in inches.



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Custodians:
Army - AR
Air Force - 11

Review activities:
Army - AT, AV, CE, EL, GL, ME, MI
Navy - MC, SH
Air Force - 99
DLA - IS
NS

User activities:
Navy - OS, YD

Preparing activity:
Air Force - 11

Agent:
DLA - IS

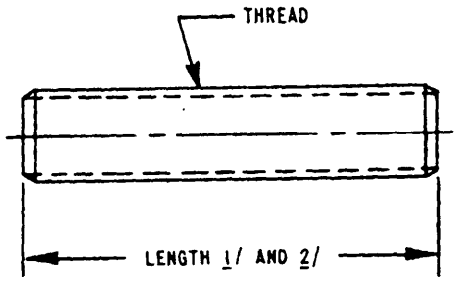
(Project 5307-0227)



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SECTION 101

ROD (STUD) CONTINUOUS THREAD
APPLICABLE DOCUMENT: NAS1454




| MATERIAL | PROTECTIVE FINISH |
|---|---|
| STEEL, CARBON | CADMIUM PLATE |
| CRES | PASSIVATE |

TABLE I

| THREAD SIZE | NAS1454 |
|---|---|
| 4-40UNC-2A | -04 |
| 6-32UNC-2A | -06 |
| 8-32UNC-2A | -08 |
| 10-32UNF-2A | - 3 |
| 1/4-28UNF-2A | - 4 |
| 5/16-24UNF-2A | - 5 |
| 3/8-24UNF-2A | - 6 |
| 1/2-20UNF-2A | - 8 |
| 5/8-18UNF-2A | -10 |
| 3/4-16UNF-2A | -12 |
| 1-12UNF-2A | -16 |

1/ PREFERRED LENGTHS TO BE IN:
1/8 INCH INCREMENTS UP TO AND INCLUDING 2 INCHES
1/4 INCH INCREMENTS FROM 2-1/4 UP TO AND INCLUDING 6 INCHES
1/2 INCH INCREMENTS FROM 6-1/2 UP TO AND INCLUDING 12 INCHES

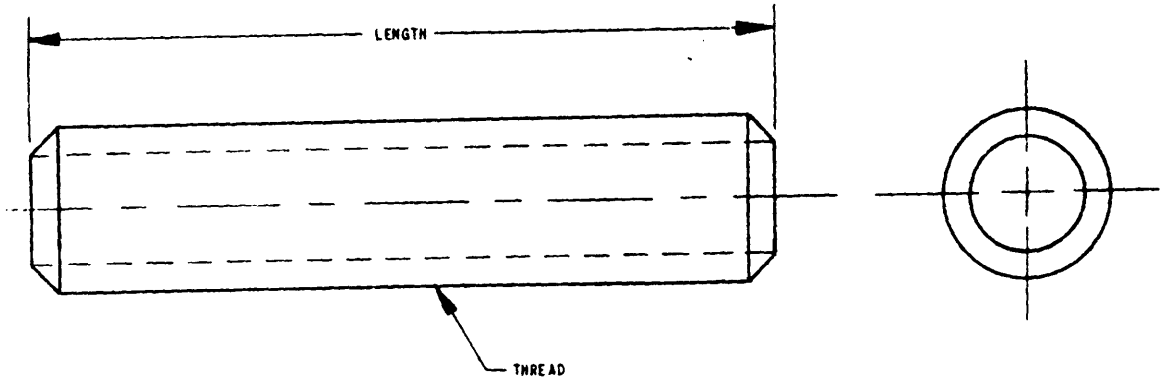
2/ LENGTHS ARE NOT TO EXCEED 12 INCHES SO AS TO CONFORM TO THE CONTINUOUS STUD DESCRIPTION



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SECTION 102

STUD, CONTINUOUS THREAD, HIGH TEMPERATURE SERVICE (875°-1000°F)
APPLICABLE DOCUMENT: MS16187




| MATERIAL | TENSILE STRENGTH (PSI) MIN | PROTECTIVE FINISH |
|---|---|---|
| STEEL, ALLOY GRADE 815 | 125,000 | NONE |



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TABLE I TABULATED LENGTHS

| LENGTH | MS16187 (FOR DASH NOS. SEE BELOW) | | | | | | |
|---|---|---|---|---|---|---|---|
| | THREAD SIZE | | | | | | |
| | 1/4-20UNC-2A | 5/16-18UNC-2A | 3/8-16UNC-2A | 1/2-13UNC-2A | 5/8-11UNC-2A | 3/4-10UNC-2A | 1-8UNC-2A |
| 1 | -1 | -14 | -27 | — | — | — | — |
| 1-1/4 | -2 | -15 | -28 | — | — | — | — |
| 1-1/2 | -3 | -16 | -29 | -57 | — | — | — |
| 1-3/4 | -4 | -17 | -30 | -58 | — | — | — |
| 2 | -5 | -18 | -31 | -59 | — | — | — |
| 2-1/4 | -6 | -19 | -32 | -60 | — | — | — |
| 2-1/2 | -7 | -20 | -33 | -61 | -80 | — | — |
| 2-3/4 | -8 | -21 | -34 | -62 | -81 | — | — |
| 3 | -9 | -22 | -35 | -63 | -82 | -99 | — |
| 3-1/4 | -10 | -23 | -36 | -64 | -83 | -100 | — |
| 3-1/2 | -11 | -24 | -37 | -65 | -84 | -101 | — |
| 3-3/4 | -12 | -25 | -38 | -66 | -85 | -102 | — |
| 4 | -13 | -26 | -39 | -67 | -86 | -103 | — |
| 4-1/4 | — | — | — | -68 | -87 | -104 | — |
| 4-1/2 | — | — | — | -69 | -88 | -105 | -129 |
| 4-3/4 | — | — | — | -70 | -89 | -106 | -130 |
| 5 | — | — | — | -71 | -90 | -107 | -131 |
| 5-1/2 | — | — | — | -72 | -91 | -108 | -132 |
| 6 | — | — | — | -73 | -92 | -109 | -133 |
| 6-1/2 | — | — | — | -74 | -93 | -110 | -134 |
| 7 | — | — | — | -75 | -94 | -111 | -135 |
| 7-1/2 | — | — | — | -76 | -95 | -112 | -136 |
| 8 | — | — | — | -77 | -96 | -113 | -137 |
| 8-1/2 | — | — | — | -78 | -97 | -114 | -138 |
| 9 | — | — | — | -79 | -98 | -115 | -139 |

TABLE I TABULATED LENGTHS (CONTINUED)

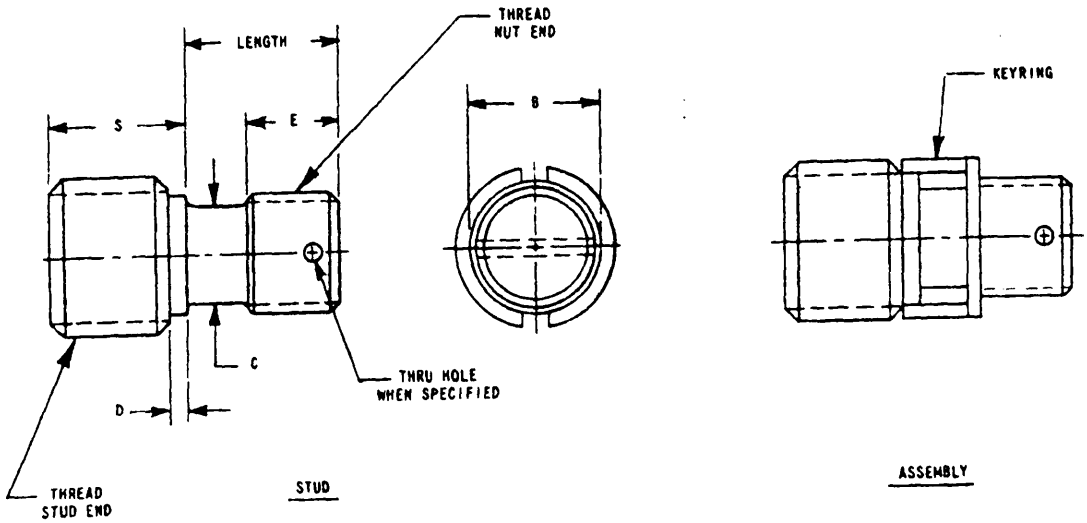
| LENGTH | MS16187 (FOR DASH NOS. SEE BELOW) | | | | | |
|---|---|---|---|---|---|---|
| | THREAD SIZE | | | | | |
| | 1-1/4-8UN-2A | 1-1/2-8UN-2A | 1-3/4-8UN-2A | 2-8UN-2A | 2-1/4-8UN-2A | 2-1/2-8UN-2A |
| 5 | — | -166 | — | — | — | — |
| 5-1/2 | — | -167 | — | — | — | — |
| 6 | -148 | -168 | -177 | -189 | -201 | -213 |
| 6-1/2 | -149 | -169 | -178 | -190 | -202 | -214 |
| 7 | -150 | -170 | -179 | -191 | -203 | -215 |
| 7-1/2 | -151 | -171 | -180 | -192 | -204 | -216 |
| 8 | -152 | -172 | -181 | -193 | -205 | -217 |
| 8-1/2 | -153 | -173 | -182 | -194 | -206 | -218 |
| 9 | -154 | -174 | -183 | -195 | -207 | -219 |
| 9-1/2 | — | -175 | -184 | -196 | -208 | -220 |
| 10 | — | -176 | -185 | -197 | -209 | -221 |
| 10-1/2 | — | -249 | -186 | -198 | -210 | -222 |
| 11 | — | -250 | -187 | -199 | -211 | -223 |
| 12 | — | -251 | -188 | -200 | -212 | -224 |



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SECTION 201

STUD, KEYRING LOCKED, 125 KSI FTU, SINGLE STEP
APPLICABLE DOCUMENT: MIL-S-45933/1




| MATERIAL | TENSILE STRENGTH (PSI) MIN | PROTECTIVE FINISH |
|---|---|---|
| STEEL, ALLOY | 125,000 | CADMIUM PLATE |
| CRES | 125,000 | PASSIVATE |

TABLE I

| NUT END THREAD SIZE UNJF-3A | STUD END THREAD SIZE UNJF-3A | B | C | D | E | S | MS45933/1 |
|---|---|---|---|---|---|---|---|
| .1900-32 | .2500-28 | .185 | .171 | .037 | .437 | .312 | -4 |
| .2500-28 | .3125-24 | .244 | .228 | .040 | .500 | .437 | -5 |
| .3125-24 | .3750-24 | .306 | .287 | .045 | .562 | .500 | -6 |
| .3750-24 | .4375-20 | .369 | .349 | .052 | .625 | .687 | -7 |



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TABLE II TABULATED LENGTHS

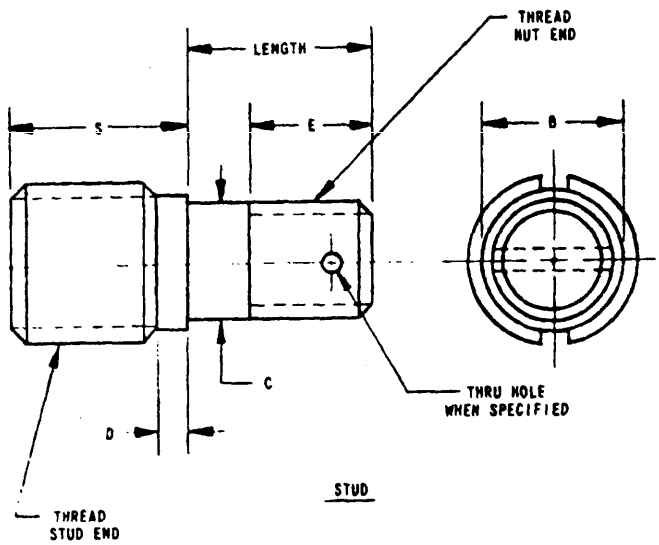
| LENGTH NUT END | LENGTH DASH NUMBER | | | |
|---|---|---|---|---|
| | NOMINAL DIA (NUT END) | | | |
| | .1900 | .2500 | .3125 | .3750 |
| .375 | -6 | -6 | — | — |
| .437 | -7 | -7 | — | — |
| .500 | -8 | -8 | — | — |
| .562 | -9 | -9 | -9 | — |
| .625 | -10 | -10 | -10 | -10 |
| .687 | -11 | -11 | -11 | -11 |
| .750 | -12 | -12 | -12 | -12 |
| .875 | -14 | -14 | -14 | -14 |
| 1.000 | -16 | -16 | -16 | -16 |
| 1.125 | -18 | -18 | -18 | -18 |
| 1.250 | -20 | -20 | -20 | -20 |
| 1.375 | -22 | -22 | -22 | -22 |
| 1.500 | -24 | -24 | -24 | -24 |
| 1.625 | -26 | -26 | -26 | -26 |
| 1.750 | -28 | -28 | -28 | -28 |
| 1.875 | -30 | -30 | -30 | -30 |
| 2.000 | -32 | -32 | -32 | -32 |
| 2.250 | — | -36 | -36 | -36 |
| 2.500 | — | -40 | -40 | -40 |
| 2.750 | — | -44 | -44 | -44 |
| 3.000 | — | -48 | -48 | -48 |
| 3.250 | — | -52 | -52 | -52 |
| 3.500 | — | -56 | -56 | -56 |
| 3.750 | — | -60 | -60 | -60 |
| 4.000 | — | -64 | -64 | -64 |



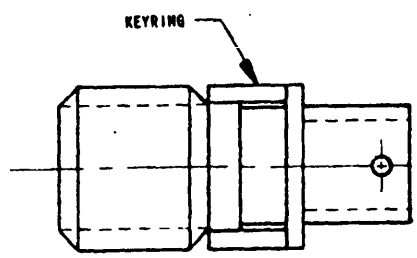
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SECTION 202

STUD, KEYRING LOCKED, 160 KSI FTU, DOUBLE STEP
APPLICABLE DOCUMENT: MIL-S-45933/2




STUD




ASSEMBLY

| MATERIAL | TENSILE STRENGTH (PSI) MIN | PROTECTIVE FINISH |
|---|---|---|
| STEEL, ALLOY | 160,000 | CADMIUM PLATE |
| CRES | 160,000 | PASSIVATE |



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TABLE I

| NUT END THREAD SIZE UNJF-3A 1/ | STUD END THREAD SIZE UNJF-3A 1/ | B | C | D | E | S | M45933/2 |
|---|---|---|---|---|---|---|---|
| .1120-40 | .1640-32 | .122 | .112 | .025 | .343 | .190 | -1 1/ |
| .1380-32 | .1900-32 | .130 | .135 | .032 | .375 | .250 | -2 1/ |
| .1640-32 | .2500-28 | .179 | .164 | .037 | .406 | .250 | -3 1/ |
| .1900-32 | .3125-24 | .236 | .190 | .040 | .437 | .312 | -4 |
| .2500-28 | .3750-24 | .289 | .250 | .045 | .500 | .437 | -5 |
| .3125-24 | .4375-20 | .338 | .312 | .052 | .562 | .562 | -6 |
| .3750-24 | .5625-18 | .447 | .375 | .060 | .625 | .625 | -7 |
| .4375-20 | .6250-18 | .507 | .437 | .060 | .687 | .750 | -8 |
| .5000-20 | .7500-16 | .619 | .500 | .067 | .812 | .875 | -9 |
| .5625-18 | .8750-14 | .728 | .562 | .075 | .875 | .937 | -10 |
| .6250-18 | 1.0000-12 | .836 | .625 | .083 | .937 | 1.000 | -11 |
| .7500-16 | 1.1250-12 | .930 | .750 | .099 | 1.062 | 1.250 | -12 |
| 1.0000-12 | 1.5000-12 | 1.273 | 1.000 | .114 | 1.312 | 1.625 | -16 |

1/ DASH 1, -2 AND -3 NUT END THREADS ARE UNJC-3A. DASH 1 STUD END IS UNJC-3A.

TABLE II TABULATED LENGTHS

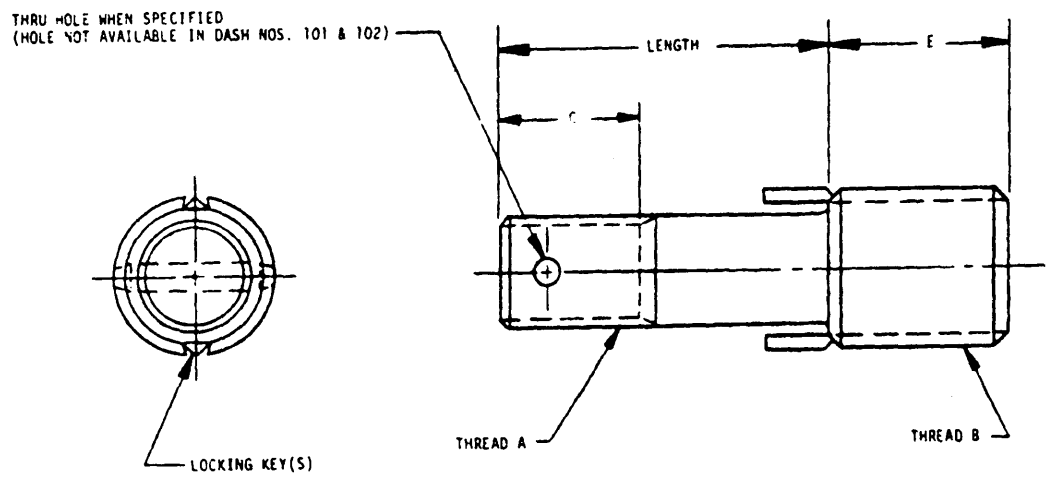
| LENGTH NUT END | LENGTH DASH NUMBER | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | NOMINAL DIA (NUT END) | | | | | | | | | | | | |
| | .1120 | .1380 | .1640 | .1900 | .2500 | .3125 | .3750 | .4375 | .5000 | .5625 | .6250 | .7500 | 1.000 |
| .250 | -4 | -4 | — | — | — | — | — | — | — | — | — | — | — |
| .312 | -5 | -5 | — | — | — | — | — | — | — | — | — | — | — |
| .375 | -6 | -6 | -6 | -6 | -6 | — | — | — | — | — | — | — | — |
| .437 | -7 | -7 | -7 | -7 | -7 | — | — | — | — | — | — | — | — |
| .500 | -8 | -8 | -8 | -8 | -8 | — | — | — | — | — | — | — | — |
| .562 | -9 | -9 | -9 | -9 | -9 | -9 | — | — | — | — | — | — | — |
| .625 | -10 | -10 | -10 | -10 | -10 | -10 | -10 | -10 | — | — | — | — | — |
| .687 | — | — | -11 | -11 | -11 | -11 | -11 | -11 | — | — | — | — | — |
| .750 | — | — | -12 | -12 | -12 | -12 | -12 | -12 | -12 | — | — | — | — |
| 1.000 | — | — | -16 | -16 | -16 | -16 | -16 | -16 | -16 | -16 | -16 | -16 | — |
| 1.125 | — | — | -18 | -18 | -18 | -18 | -18 | -18 | -18 | -18 | -18 | -18 | -18 |
| 1.250 | — | — | -20 | -20 | -20 | -20 | -20 | -20 | -20 | -20 | -20 | -20 | -20 |
| 1.375 | — | — | -22 | -22 | -22 | -22 | -22 | -22 | -22 | -22 | -22 | -22 | -22 |
| 1.500 | — | — | — | -24 | -24 | -24 | -24 | -24 | -24 | -24 | -24 | -24 | -24 |
| 1.625 | — | — | — | -26 | -26 | -26 | -26 | -26 | -26 | -26 | -26 | -26 | -26 |
| 1.750 | — | — | — | -28 | -28 | -28 | -28 | -28 | -28 | -28 | -28 | -28 | -28 |
| 1.875 | — | — | — | -30 | -30 | -30 | -30 | -30 | -30 | -30 | -30 | -30 | -30 |
| 2.000 | — | — | — | -32 | -32 | -32 | -32 | -32 | -32 | -32 | -32 | -32 | -32 |
| 2.250 | — | — | — | — | -36 | -36 | -36 | -36 | -36 | -36 | -36 | -36 | -36 |
| 2.500 | — | — | — | — | -40 | -40 | -40 | -40 | -40 | -40 | -40 | -40 | -40 |
| 2.750 | — | — | — | — | -44 | -44 | -44 | -44 | -44 | -44 | -44 | -44 | -44 |
| 3.000 | — | — | — | — | -48 | -48 | -48 | -48 | -48 | -48 | -48 | -48 | -48 |
| 3.250 | — | — | — | — | -52 | -52 | -52 | -52 | -52 | -52 | -52 | -52 | -52 |
| 3.500 | — | — | — | — | -56 | -56 | -56 | -56 | -56 | -56 | -56 | -56 | -56 |
| 3.750 | — | — | — | — | -60 | -60 | -60 | -60 | -60 | -60 | -60 | -60 | -60 |
| 4.000 | — | — | — | — | -64 | -64 | -64 | -64 | -64 | -64 | -64 | -64 | -64 |



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SECTION 203

STUD, LOCKED-IN, KEY LOCKED HEAVY DUTY
APPLICABLE DOCUMENT: MS51834




| MATERIAL | TENSILE STRENGTH (PSI) MIN | PROTECTIVE FINISH |
|---|---|---|
| STEEL, ALLOY | 160,000 | CADMIUM PLATE |
| CRES, A286 | 140,000 | PASSIVATE |



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TABLE I (COARSE THREAD NUT END)

| A NUT END THREAD SIZE UNJC-3A | B STUD END THREAD SIZE MOD MINOR DIA | C 1/ | E | MS51834 |
|---|---|---|---|---|
| .164-32 | .3125-18UNC-2A | .410 | .312 | -101 |
| .190-24 | .375-16UNC-2A | .410 | .312 | -102 |
| .250-20 | .4375-14UNC-2A | .470 | .375 | -103 |
| .3125-18 | .500-13UNC-2A | .530 | .437 | -104 |
| .375-16 | .5625-12UNC-2A | .640 | .500 | -105 |
| .4375-14 | .625-11UNC-2A | .660 | .625 | -106 |
| .500-13 | .6875-11UNS-2A | .780 | .688 | -107 |
| .625-11 | .875-14UNF-2A | .950 | .875 | -109 |
| .750-10 | 1.125-12UNF-2A | 1.090 | 1.125 | -110 |
| .875-9 | 1.250-12UNF-2A | 1.250 | 1.250 | -111 |
| 1.000-8 | 1.375-12UNF-2A | 1.370 | 1.375 | -112 |

TABLE II (FINE THREAD NUT END)

| A NUT END THREAD SIZE UNJF-3A | B STUD END THREAD SIZE MOD MINOR DIA | C | E | MS51834 |
|---|---|---|---|---|
| .164-36 | .3125-18UNC-2A | .410 | .312 | -201 |
| .190-32 | .375-16UNC-2A | .410 | .312 | -202 |
| .250-28 | .4375-14UNC-2A | .470 | .375 | -203 |
| .3125-24 | .500-13UNC-2A | .530 | .437 | -204 |
| .375-24 | .5625-12UNC-2A | .640 | .500 | -205 |
| .4375-20 | .625-11UNC-2A | .660 | .625 | -206 |
| .500-20 | .6875-11UNS-2A | .780 | .688 | -207 |
| .625-18 | .875-14UNF-2A | .950 | .875 | -209 |
| .750-16 | 1.125-12UNF-2A | 1.090 | 1.125 | -210 |
| .875-14 | 1.250-12UNF-2A | 1.250 | 1.250 | -211 |
| 1.000-12 | 1.375-12UNF-2A | 1.370 | 1.375 | -212 |

1/ THIS DIMENSION DOES NOT APPLY TO DASH NUMBERS IN TABLE III ABOVE HEAVY LINES.

TABLE III - TABULATED LENGTHS

| LENGTH NUT END | LENGTH DASH NUMBER 2/ (TABLE APPLICABLE TO TABLES I AND II) NOMINAL DIA (NUT END) | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | .164 | .190 | .250 | .3125 | .375 | .4375 | .500 | .625 | .750 | .875 | 1.000 |
| .312 | -5 | — | — | — | — | — | — | — | — | — | — |
| .375 | -6 | -6 | -6 | — | — | — | — | — | — | — | — |
| .438 | -7 | -7 | -7 | — | — | — | — | — | — | — | — |
| .500 | -8 | -8 | -8 | -8 | — | — | — | — | — | — | — |
| .562 | -9 | -9 | -9 | -9 | -9 | — | — | — | — | — | — |
| .625 | -10 | -10 | -10 | -10 | -10 | -10 | — | — | — | — | — |
| .750 | -12 | -12 | -12 | -12 | -12 | -12 | -12 | — | — | — | — |
| .875 | -14 | -14 | -14 | -14 | -14 | -14 | -14 | — | — | — | — |
| 1.000 | -16 | -16 | -16 | -16 | -16 | -16 | -16 | -16 | — | — | — |
| 1.125 | -18 | -18 | -18 | -18 | -18 | -18 | -18 | -18 | -18 | — | — |
| 1.250 | -20 | -20 | -20 | -20 | -20 | -20 | -20 | -20 | -20 | — | — |
| 1.375 | -22 | -22 | -22 | -22 | -22 | -22 | -22 | -22 | -22 | -22 | -22 |
| 1.500 | -24 | -24 | -24 | -24 | -24 | -24 | -24 | -24 | -24 | -24 | -24 |
| 1.625 | -26 | -26 | -26 | -26 | -26 | -26 | -26 | -26 | -26 | -26 | -26 |
| 1.750 | -28 | -28 | -28 | -28 | -28 | -28 | -28 | -28 | -28 | -28 | -28 |
| 1.875 | -30 | -30 | -30 | -30 | -30 | -30 | -30 | -30 | -30 | -30 | -30 |
| 2.000 | -32 | -32 | -32 | -32 | -32 | -32 | -32 | -32 | -32 | -32 | -32 |
| 2.250 | — | -36 | -36 | -36 | -36 | -36 | -36 | -36 | -36 | -36 | -36 |
| 2.500 | — | -40 | -40 | -40 | -40 | -40 | -40 | -40 | -40 | -40 | -40 |
| 2.750 | — | -44 | -44 | -44 | -44 | -44 | -44 | -44 | -44 | -44 | -44 |
| 3.000 | — | -48 | -48 | -48 | -48 | -48 | -48 | -48 | -48 | -48 | -48 |
| 3.250 | — | -52 | -52 | -52 | -52 | — | -52 | — | -52 | -52 | -52 |
| 3.500 | — | -56 | -56 | -56 | -56 | — | -56 | — | -56 | -56 | -56 |
| 3.750 | — | -60 | -60 | -60 | -60 | — | -60 | — | -60 | -60 | -60 |
| 4.000 | — | — | -64 | -64 | -64 | — | -64 | — | -64 | -64 | -64 |

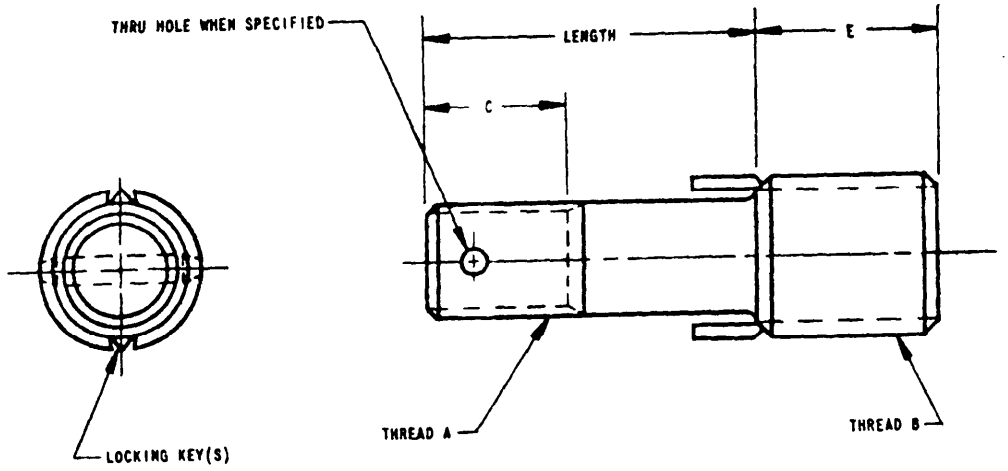
2/ DASH NUMBERED PARTS ABOVE THE HEAVY LINE ARE THREADED TO THE STUD END. THE MAXIMUM DISTANCE FROM THE STUD END TO THE FIRST FULL THREAD IS EQUAL TO THE SUM OF THE MAXIMUM FILLET RADIUS (.075R) AND A MAXIMUM OF TWO INCOMPLETE THREADS.



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SECTION 204

STUD, LOCKED IN, KEY LOCKED, LIGHTWEIGHT
(APPLICABLE DOCUMENT: MS51833)




| MATERIAL | TENSILE STRENGTH (PSI) MIN | PROTECTIVE FINISH |
|---|---|---|
| STEEL, ALLOY | 160,000 | CADMIUM PLATE |
| CRES, A-286 | 140,000 | PASSIVATE |



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TABLE I (COARSE THREAD NUT END)

| A NUT END THREAD SIZE UNJC-3A | B STUD END THREAD SIZE MOD MINOR DIA | C 1/ | E | MS51833 |
|---|---|---|---|---|
| .190-24 | .3125-18UNC-2A | .410 | .312 | -101 |
| .250-20 | .375-16UNC-2A | .470 | .375 | -102 |
| .3125-18 | .4375-14UNC-2A | .530 | .437 | -103 |
| .375-16 | .500-13UNC-2A | .640 | .500 | -104 |
| .500-13 | .625-11UNC-2A | .780 | .625 | -106 |

TABLE II (FINE THREAD NUT END)

| A NUT END THREAD SIZE UNJF-3A | B STUD END THREAD SIZE MOD MINOR DIA | C | E | MS51833 |
|---|---|---|---|---|
| .190-32 | .3125-18UNC-2A | .410 | .312 | -201 |
| .250-28 | .375-16UNC-2A | .470 | .375 | -202 |
| .3125-24 | .4375-14UNC-2A | .530 | .437 | -203 |
| .375-24 | .500-13UNC-2A | .640 | .500 | -204 |
| .500-20 | .625-11UNC-2A | .780 | .625 | -206 |

1/ THIS DIMENSION DOES NOT APPLY TO DASH NUMBER IN TABLE ABOVE HEAVY LINES.

TABLE III (TABULATED LENGTHS)

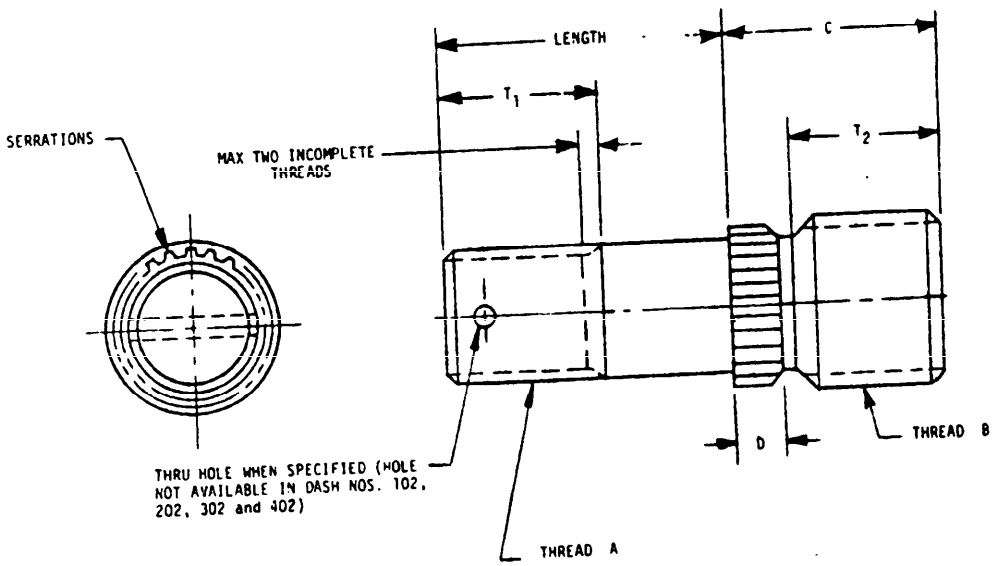
| LENGTH NUT END | LENGTH DASH NUMBER (TABLE APPLICABLE TO TABLES I AND II) | | | | |
|---|---|---|---|---|---|
| | NOMINAL DIA (NUT END) | | | | |
| | .190 | .250 | .3125 | .375 | .500 |
| .375 | -6 | -6 | — | — | — |
| .438 | -7 | -7 | — | — | — |
| .500 | -8 | -8 | — | — | — |
| .562 | -9 | -9 | -9 | — | — |
| .625 | -10 | -10 | -10 | -10 | — |
| .750 | -12 | -12 | -12 | -12 | -12 |
| .875 | -14 | -14 | -14 | -14 | -14 |
| 1.000 | -16 | -16 | -16 | -16 | -16 |
| 1.125 | -18 | -18 | -18 | -18 | -18 |
| 1.188 | -19 | -19 | -19 | -19 | -19 |
| 1.250 | -20 | -20 | -20 | -20 | -20 |
| 1.375 | -22 | -22 | -22 | -22 | -22 |
| 1.500 | -24 | -24 | -24 | -24 | -24 |
| 1.625 | -26 | -26 | -26 | -26 | -26 |
| 1.750 | -28 | -28 | -28 | -28 | -28 |
| 1.875 | -30 | -30 | -30 | -30 | -30 |
| 2.000 | -32 | -32 | -32 | -32 | -32 |
| 2.250 | -36 | -36 | -36 | -36 | -36 |
| 2.500 | -40 | -40 | -40 | -40 | -40 |
| 2.750 | — | -44 | -44 | -44 | -44 |
| 3.000 | — | -48 | -48 | -48 | -48 |
| 3.250 | — | -52 | -52 | -52 | -52 |
| 3.500 | — | -56 | -56 | -56 | -56 |
| 3.750 | — | -60 | -60 | -60 | -60 |
| 4.000 | — | -64 | -64 | -64 | -64 |



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SECTION 205

STUD, LOCKED IN, RING LOCKED, SERRATED
APPLICABLE DOCUMENT: MS51989




| MATERIAL | TENSILE STRENGTH (PSI) MIN | PROTECTIVE FINISH |
|---|---|---|
| STEEL, ALLOY | 125,000 | CADMIUM PLATE |
| CRES | 125,000 | PASSIVATE |



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TABLE I (FINE THD NUT END-COARSE THD STUD END)

| A NUT END THREAD SIZE UNF-3A | B STUD END THREAD SIZE (COARSE) SEE NOTE 1/ | C | D | $T_1$ | $T_2$ | MS51989 |
|---|---|---|---|---|---|---|
| .138-40 | .164-32 | .250 | .060 | .380 | .140 | -102 |
| .164-36 | .190-24 | .380 | .080 | .410 | .230 | -103 |
| .190-32 | .250-20 | .440 | .080 | .440 | .280 | -104 |
| .250-28 | .3125-18 | .560 | .080 | .500 | .390 | -105 |
| .3125-24 | .375-16 | .690 | .080 | .560 | .510 | -106 |
| .375-24 | .4375-14 | .750 | .120 | .620 | .520 | -107 |
| .4375-20 | .500-13 | .810 | .120 | .690 | .570 | -108 |
| .500-20 | .5625-12 | .880 | .120 | .810 | .630 | -109 |
| .5625-18 | .625-11 | 1.000 | .140 | .940 | .720 | -110 |
| .625-18 | .750-10 | 1.120 | .160 | 1.000 | .820 | -111 |
| .750-16 | .875-9 | 1.310 | .160 | 1.120 | .990 | -112 |

TABLE II (FINE THD NUT END-FINE THD STUD END)

| A NUT END THREAD SIZE UNF-3A | B STUD END THREAD SIZE (FINE) SEE NOTE 1/ | C | D | $T_1$ | $T_2$ | MS51989 |
|---|---|---|---|---|---|---|
| .138-30 | .164-36 | .250 | .060 | .380 | .140 | -202 |
| .164-36 | .190-32 | .380 | .080 | .410 | .250 | -203 |
| .190-32 | .250-28 | .440 | .080 | .440 | .310 | -204 |
| .250-28 | .3125-24 | .560 | .080 | .500 | .420 | -205 |
| .3125-24 | .375-24 | .690 | .080 | .560 | .550 | -206 |
| .375-24 | .4375-20 | .750 | .120 | .620 | .550 | -207 |
| .4375-20 | .500-20 | .810 | .120 | .690 | .610 | -208 |
| .500-20 | .5625-18 | .880 | .120 | .810 | .670 | -209 |
| .5625-18 | .625-18 | 1.000 | .140 | .940 | .780 | -210 |
| .625-18 | .750-16 | 1.120 | .160 | 1.000 | .880 | -211 |
| .750-16 | .875-14 | 1.310 | .160 | 1.120 | 1.050 | -212 |

TABLE III (COARSE THD NUT END-COARSE THD STUD END)

| A NUT END THREAD SIZE UNC-3A | B STUD END THREAD SIZE (COARSE) SEE NOTE 1/ | C | D | $T_1$ | $T_2$ | MS51989 |
|---|---|---|---|---|---|---|
| .138-32 | .164-32 | .250 | .060 | .380 | .140 | -302 |
| .164-32 | .190-24 | .380 | .080 | .410 | .230 | -303 |
| .190-24 | .250-20 | .440 | .080 | .440 | .280 | -304 |
| .250-20 | .3125-18 | .560 | .080 | .500 | .390 | -305 |
| .3125-18 | .375-16 | .690 | .080 | .560 | .510 | -306 |
| .375-16 | .4375-14 | .750 | .120 | .620 | .520 | -307 |
| .4375-14 | .500-13 | .810 | .120 | .690 | .570 | -308 |
| .500-13 | .5625-12 | .880 | .120 | .810 | .630 | -309 |
| .5625-12 | .625-11 | 1.000 | .140 | .940 | .720 | -310 |
| .625-11 | .750-10 | 1.120 | .160 | 1.000 | .820 | -311 |
| .750-10 | .875-9 | 1.310 | .160 | 1.120 | .990 | -312 |

TABLE IV (COARSE THD NUT END-FINE THD STUD END)

| A NUT END THREAD SIZE UNC-3A | B STUD END THREAD SIZE (FINE) SEE NOTE 1/ | C | D | $T_1$ | $T_2$ | MS51989 |
|---|---|---|---|---|---|---|
| .138-32 | .164-36 | .250 | .060 | .380 | .140 | -402 |
| .164-32 | .190-32 | .380 | .080 | .410 | .250 | -403 |
| .190-24 | .250-28 | .440 | .080 | .440 | .310 | -404 |
| .250-20 | .3125-24 | .560 | .080 | .500 | .420 | -405 |
| .3125-18 | .375-24 | .690 | .080 | .560 | .550 | -406 |
| .375-16 | .4375-20 | .750 | .120 | .620 | .550 | -407 |
| .4375-14 | .500-20 | .810 | .120 | .690 | .610 | -408 |
| .500-13 | .5625-18 | .880 | .120 | .810 | .670 | -409 |
| .5625-12 | .625-18 | 1.000 | .140 | .940 | .780 | -410 |
| .625-11 | .750-16 | 1.120 | .160 | 1.000 | .880 | -411 |
| .750-10 | .375-14 | 1.310 | .160 | 1.120 | 1.050 | -412 |

1/ THE STUD END THREAD HAS A SPECIAL PITCH DIAMETER AND MINOR DIAMETER WHICH INSTALLS INTO A NATIONAL CLASS 3 TAPPED HOLE.



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TABLE V
TABULATED LENGTHS (NUT END)

| LENGTH NUT END | LENGTH DASH NUMBER 1/ (TABLE APPLICABLE TO TABLES I, II, III, AND IV) | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | NOMINAL DIA (NUT END) | | | | | | | | | | |
| | .138 | .164 | .190 | .250 | .3125 | .375 | .4375 | .500 | .5625 | .625 | .750 |
| .250 | -4 | — | — | — | — | — | — | — | — | — | — |
| .312 | -5 | -5 | — | — | — | — | — | — | — | — | — |
| .375 | -6 | -6 | -6 | -6 | — | — | — | — | — | — | — |
| .438 | -7 | -7 | -7 | -7 | — | — | — | — | — | — | — |
| .500 | -8 | -8 | -8 | -8 | -8 | — | — | — | — | — | — |
| .562 | -9 | -9 | -9 | -9 | -9 | -9 | — | — | — | — | — |
| .625 | -10 | -10 | -10 | -10 | -10 | -10 | -10 | — | — | — | — |
| .688 | — | — | -11 | -11 | -11 | — | — | — | — | — | — |
| .750 | -12 | -12 | -12 | -12 | -12 | -12 | -12 | -12 | — | — | — |
| .875 | -14 | -14 | -14 | -14 | -14 | -14 | -14 | -14 | -14 | — | — |
| 1.000 | -16 | -16 | -16 | -16 | -16 | -16 | -16 | -16 | -16 | -16 | — |
| 1.125 | -18 | -18 | -18 | -18 | -18 | -18 | -18 | -18 | -18 | -18 | -18 |
| 1.250 | -20 | -20 | -20 | -20 | -20 | -20 | -20 | -20 | -20 | -20 | -20 |
| 1.375 | -22 | -22 | -22 | -22 | -22 | -22 | -22 | -22 | -22 | -22 | -22 |
| 1.500 | — | -24 | -24 | -24 | -24 | -24 | -24 | -24 | -24 | -24 | -24 |
| 1.625 | — | -26 | -26 | -26 | -26 | -26 | -26 | -26 | -26 | -26 | -26 |
| 1.750 | — | -28 | -28 | -28 | -28 | -28 | -28 | -28 | -28 | -28 | -28 |
| 1.875 | — | -30 | -30 | -30 | -30 | -30 | -30 | -30 | -30 | -30 | -30 |
| 2.000 | — | -32 | -32 | -32 | -32 | -32 | -32 | -32 | -32 | -32 | -32 |
| 2.250 | — | — | -36 | -36 | -36 | -36 | -36 | -36 | -36 | -36 | -36 |
| 2.500 | — | — | -40 | -40 | -40 | -40 | — | -40 | -40 | — | -40 |
| 2.750 | — | — | -44 | -44 | -44 | -44 | — | -44 | -44 | — | -44 |
| 3.000 | — | — | -48 | -48 | -48 | -48 | — | -48 | -48 | — | -48 |
| 3.250 | — | — | -52 | -52 | -52 | -52 | — | -52 | -52 | — | — |
| 3.500 | — | — | -56 | -56 | -56 | -56 | — | -56 | -56 | — | — |
| 3.750 | — | — | -60 | -60 | -60 | -60 | — | -60 | -60 | — | — |
| 4.000 | — | — | — | -64 | -64 | -64 | — | — | — | — | — |

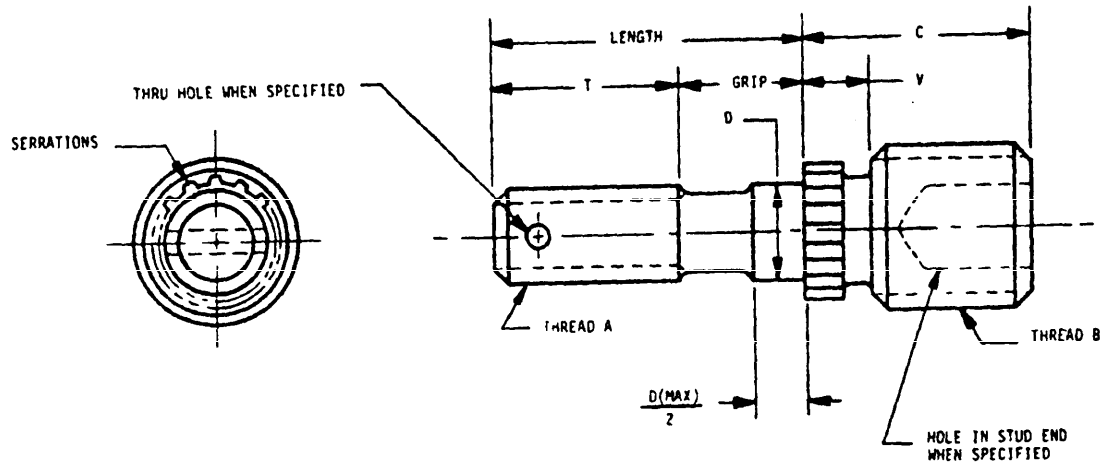
1/ DASH NUMBERED PARTS ABOVE THE HEAVY LINE ARE THREADED TO THE SERRATED COLLAR. THE MAXIMUM DISTANCE FROM THE SERRATED COLLAR TO THE FIRST FULL THREAD IS EQUAL TO THE SUM OF THE MAXIMUM FILLET RADIUS (.015R) AND A MAXIMUM OF TWO INCOMPLETE THREADS.



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SECTION 206

STUD, LOCKED IN, RING LOCKED, SERRATED, HIGH STRENGTH
APPLICABLE DOCUMENT: MS51992




| MATERIAL | TENSILE STRENGTH (PSI) MIN | PROTECTIVE FINISH | MATERIAL CODE LETTER | MS51992 APPLICABLE DASH NUMBERS |
|---|---|---|---|---|
| STEEL, ALLOY | 160,000 | CADMIUM PLATE | A | -642 THRU -648 & -802 THRU -808 |
| STEEL, ALLOY | 180,000 | CADMIUM PLATE | B | -502 THRU -508 |
| CRES | 140,000 | PASSIVATE | C | -642 THRU -648 |
| NICKEL BASE ALLOY | 180,000 | NONE | D | -502 THRU -508 |
| TITANIUM, ALLOY | 160,000 | NONE | E | -642 THRU -648 -802 THRU -808 |



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TABLE I (FINE THREAD NUT END AND FINE THREAD STUD END-SHORT LENGTH)

| A NUT END THREAD SIZE UNJF-3A | B STUD END THREAD SIZE SEE NOTE 1/ | C | D DIA MAX | T | V | MS51992 |
|---|---|---|---|---|---|---|
| .1900-32 | .3125-24 | .386 | .189 | .469 | .128 | -502 |
| .2500-28 | .3750-24 | .486 | .249 | .594 | .142 | -503 |
| .3125-24 | .4375-20 | .627 | .312 | .688 | .176 | -504 |
| .3750-24 | .5000-20 | .762 | .374 | .750 | .200 | -505 |
| .4375-20 | .6250-18 | .808 | .437 | .812 | .200 | -506 |
| .5000-20 | .7500-16 | .883 | .499 | .875 | .200 | -507 |
| .6250-18 | .8750-14 | 1.153 | .624 | 1.000 | .255 | -508 |

TABLE II (FINE THREAD NUT END AND FINE THREAD STUD END-MEDIUM LENGTH)

| A NUT END THREAD SIZE UNJF-3A | B STUD END THREAD SIZE SEE NOTE 1/ | C | D DIA MAX | T | V | MS51992 |
|---|---|---|---|---|---|---|
| .1900-32 | .3125-24 | .433 | .189 | .469 | .128 | -642 |
| .2500-28 | .3750-24 | .558 | .249 | .594 | .142 | -643 |
| .3125-24 | .4375-20 | .725 | .312 | .688 | .176 | -644 |
| .3750-24 | .5000-20 | .893 | .374 | .750 | .200 | -645 |
| .4375-20 | .6250-18 | .948 | .437 | .812 | .200 | -646 |
| .5000-20 | .7500-16 | 1.038 | .499 | .875 | .200 | -647 |
| .6250-18 | .8750-14 | 1.364 | .624 | 1.000 | .255 | -648 |

TABLE III (FINE THREAD NUT END AND COARSE THREAD STUD END-LONG LENGTH)

| A NUT END THREAD SIZE UNJF-3A | B STUD END THREAD SIZE SEE NOTE 1/ | C | D DIA MAX | T | V | MS51992 |
|---|---|---|---|---|---|---|
| .1900-32 | .3125-18 | .511 | .189 | .469 | .128 | -802 |
| .2500-28 | .3750-16 | .673 | .249 | .594 | .142 | -803 |
| .3125-24 | .4375-14 | .868 | .312 | .688 | .176 | -804 |
| .3750-24 | .5000-13 | 1.076 | .374 | .750 | .200 | -805 |
| .4375-20 | .6250-11 | 1.155 | .437 | .812 | .200 | -806 |
| .5000-20 | .7500-10 | 1.267 | .499 | .875 | .200 | -807 |
| .6250-18 | .8750-9 | 1.656 | .624 | 1.000 | .255 | -808 |

NOTE:
1/ THE STUD END THREAD HAS A SPECIAL PITCH DIAMETER AND MINOR DIAMETER WHICH INSTALLS INTO A CLASS 3B TAPPED HOLE.

TABLE IV TABULATED LENGTHS (NUT END)

| DASH NO. | LENGTH NUT END | GRIP LENGTH (REF) (APPLICABLE TO TABLES I, II AND III) 2/ NOMINAL DIA (NUT END) .1900 | .2500 | .3125 | .3750 | .4375 | .5000 | .6250 |
|---|---|---|---|---|---|---|---|---|
| -9 | .562 | .093 | | | | | | |
| -10 | .625 | .156 | | | | | | |
| -12 | .750 | .281 | .156 | | | | | |
| -14 | .875 | .406 | .281 | .187 | .125 | | | |
| -16 | 1.000 | .531 | .406 | .312 | .250 | .188 | | |
| -18 | 1.125 | .656 | .531 | .437 | .375 | .313 | .250 | |
| -20 | 1.250 | .781 | .656 | .562 | .500 | .438 | .375 | .250 |
| -22 | 1.375 | .906 | .781 | .687 | .625 | .563 | .500 | .375 |
| -24 | 1.500 | 1.031 | .906 | .812 | .750 | .688 | .625 | .500 |
| -26 | 1.625 | 1.156 | 1.031 | .937 | .875 | .813 | .750 | .625 |
| -28 | 1.750 | 1.281 | 1.156 | 1.062 | 1.000 | .938 | .875 | .750 |
| -30 | 1.875 | 1.406 | 1.281 | 1.187 | 1.125 | 1.063 | 1.000 | .875 |
| -32 | 2.000 | 1.531 | 1.406 | 1.312 | 1.250 | 1.188 | 1.125 | 1.000 |
| -36 | 2.250 | 1.781 | 1.656 | 1.562 | 1.500 | 1.438 | 1.375 | 1.250 |
| -40 | 2.500 | 2.031 | 1.906 | 1.812 | 1.750 | 1.688 | 1.625 | 1.500 |
| -44 | 2.750 | 2.281 | 2.156 | 2.062 | 2.000 | 1.938 | 1.875 | 1.750 |
| -48 | 3.000 | 2.531 | 2.406 | 2.312 | 2.250 | 2.188 | 2.125 | 2.000 |
| -52 | 3.250 | 2.781 | 2.656 | 2.562 | 2.500 | 2.438 | 2.375 | 2.250 |
| -56 | 3.500 | 3.031 | 2.906 | 2.812 | 2.750 | 2.688 | 2.625 | 2.500 |
| -60 | 3.750 | 3.281 | 3.156 | 3.062 | 3.000 | 2.938 | 2.875 | 2.750 |
| -64 | 4.000 | 3.531 | 3.406 | 3.312 | 3.250 | 3.188 | 3.125 | 3.000 |

2/ DASH NUMBER PARTS BELOW HEAVY LINE HAVE LENGTH OF SHANK D EQUAL TO $\frac{D(MAX)}{2}$

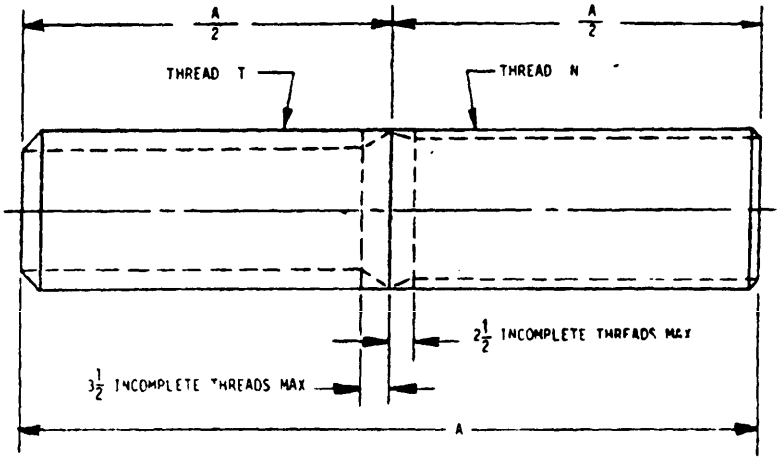
DASH NUMBER PARTS ABOVE HEAVY LINE HAVE LENGTH OF SHANK D SHORTER THAN $\frac{D(MAX)}{2}$

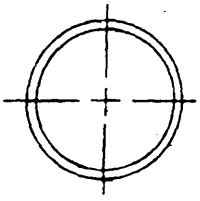


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SECTION 301

STUD, PLAIN, GENERAL PURPOSE
APPLICABLE DOCUMENT: MS51864



| MATERIAL | TENSILE STRENGTH (PSI) MIN | PROTECTIVE FINISH |
|---|---|---|
| STEEL, CARBON OR ALLOY SIZES ¼ THRU 1 | 120,000 | CADMIUM PLATE |
| OVER 1 TO 1½ | 105,000 | |
| CRES | 80,000 | PASSIVATE |
| BRASS | 60,000 | BLACK CHEMICAL FINISH |
| NICKEL-COPPER ALLOY | 80,000 | NONE |



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TABLE I - (TAP END THREAD DESIGNATION CLASS 5HF)

| T<br>TAP END<br>THREAD SIZE<br>NC-5 HF | N<br>NUT END<br>THREAD SIZE<br>UNF-2A | MS51864<br>(STEEL) |
|---|---|---|
| .2500-20 | .2500-28 | -102 |
| .3125-18 | .3125-24 | -103 |
| .3750-16 | .3750-24 | -104 |
| .5000-13 | .5000-20 | -106 |
| .6250-11 | .6250-18 | -108 |
| .7500-10 | .7500-16 | -109 |
| 1.0000- 8 | 1.0000-12 | -111 |
| 1.2500- 7 | 1.2500-12 | -113 |

TABLE II - (TAP END THREAD DESIGNATION CLASS 5 CSF)

| T<br>TAP END<br>THREAD SIZE<br>NC-5 CSF | N<br>NUT END<br>THREAD SIZE<br>UNF-2A | MS51864 | |
|---|---|---|---|
| | | (CRES) | (BRASS) |
| .2500-20 | .2500-28 | -202 | -302 |
| .3125-18 | .3125-24 | -203 | -303 |
| .3750-16 | .3750-24 | -204 | -304 |
| .5000-13 | .5000-20 | -206 | -306 |
| .6250-11 | .6250-18 | -208 | -308 |
| .7500-10 | .7500-16 | -209 | -309 |
| 1.0000- 8 | 1.0000-12 | -211 | -311 |
| 1.2500- 7 | 1.2500-12 | -213 | -313 |

TABLE III - (TAP END THREAD DESIGNATION CLASS 5 ONF)

| T<br>TAP END<br>THREAD SIZE<br>NC-5 ONF | N<br>NUT END<br>THREAD SIZE<br>UNF-2A | MS51864<br>(NI-CU) |
|---|---|---|
| .2500-20 | .2500-28 | -402 |
| .3750-16 | .3750-24 | -404 |
| .5000-13 | .5000-20 | -406 |
| .6250-11 | .6250-18 | -408 |
| .7500-10 | .7500-16 | -409 |
| 1.0000- 8 | 1.0000-12 | -411 |
| 1.2500- 7 | 1.2500-12 | -413 |



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TABLE IV TABULATED LENGTHS

| LENGTH DASH NUMBER | DIAMETER DASH NUMBERS | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 102<br>202<br>302<br>402 | 103<br>203<br>303 | 104<br>204<br>304<br>404 | 106<br>206<br>306<br>406 | 108<br>208<br>308<br>408 | 109<br>209<br>309<br>409 | 111<br>211<br>311<br>411 | 113<br>213<br>313<br>413 |
| | A | A | A | A | A | A | A | A |
| 12 | 1.500 | — | — | — | — | — | — | — |
| 14 | 1.750 | 1.750 | — | — | — | — | — | — |
| 16 | 2.000 | 2.000 | 2.000 | — | — | — | — | — |
| 18 | 2.250 | 2.250 | 2.250 | — | — | — | — | — |
| 20 | 2.500 | 2.500 | 2.500 | 2.500 | — | — | — | — |
| 22 | 2.750 | 2.750 | 2.750 | 2.750 | — | — | — | — |
| 24 | 3.000 | 3.000 | 3.000 | 3.000 | 3.000 | — | — | — |
| 26 | — | — | 3.250 | 3.250 | 3.250 | — | — | — |
| 28 | — | — | 3.500 | 3.500 | 3.500 | 3.500 | — | — |
| 30 | — | — | 3.750 | 3.750 | 3.750 | 3.750 | — | — |
| 32 | — | — | 4.000 | 4.000 | 4.000 | 4.000 | — | — |
| 34 | — | — | — | 4.250 | 4.250 | 4.250 | — | — |
| 36 | — | — | — | 4.500 | 4.500 | 4.500 | 4.500 | — |
| 38 | — | — | — | 4.750 | 4.750 | 4.750 | 4.750 | — |
| 40 | — | — | — | 5.000 | 5.000 | 5.000 | 5.000 | — |
| 44 | — | — | — | 5.500 | 5.500 | 5.500 | 5.500 | 5.500 |
| 48 | — | — | — | 6.000 | 6.000 | 6.000 | 6.000 | 6.000 |
| 52 | — | — | — | — | — | — | — | 6.500 |
| 56 | — | — | — | — | — | — | — | 7.000 |
| 60 | — | — | — | — | — | — | — | 7.500 |
| 64 | — | — | — | — | — | — | — | 8.000 |

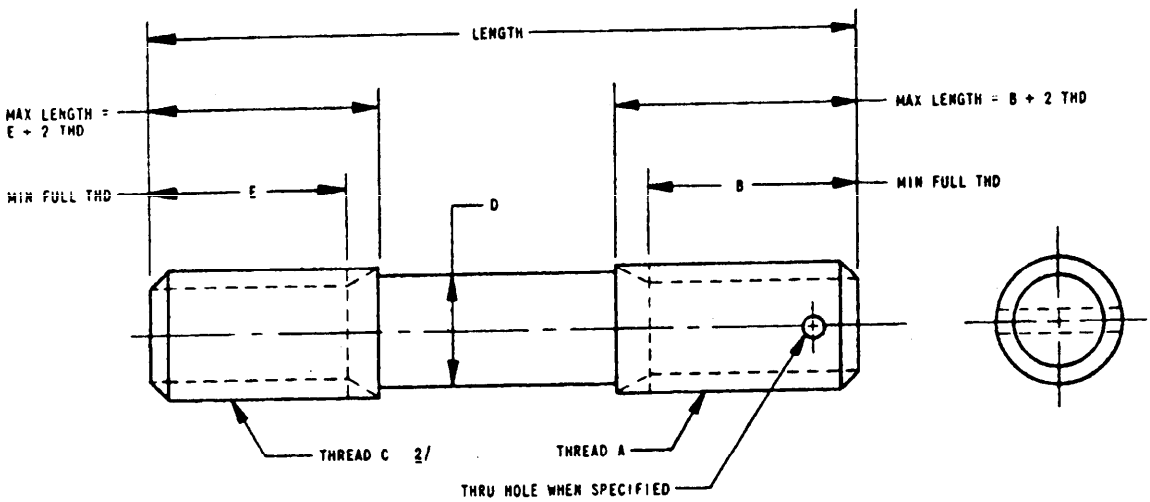


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SECTION 302

STUD, STRAIGHT

APPLICABLE DOCUMENTS: AN126587 THRU AN126652, AN128363 THRU AN128686, AN128687 THRU AN128998 MS17299, MS17300, MS17301, MS17302, MS17303




| MATERIAL | HARDNESS | PROTECTIVE FINISH |
|---|---|---|
| STEEL, ALLOY | ROCKWELL C-26-32 | CADMIUM PLATE |



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TABLE I PART NUMBERS

| A NUT END THREAD SIZE | B | C 2/ STUD END THREAD SIZE | D (NOM) | E | LENGTH (NOM) | 1/ PART NUMBER | |
|---|---|---|---|---|---|---|---|
| | | | | | | DRILLED | UNDRILLED |
| .190-32UNJF-3A | .375 | .190-24UNJS | .169 | .375 | .875 | — | MS17299-016 |
| .190-32UNJF-3A | .500 | .190-24UNJS | .169 | .375 | 1.000 | — | MS17299-026 |
| .190-32UNJF-3A | .500 | .190-24UNJS | .169 | .375 | 1.125 | — | MS17299-036 |
| .190-32UNJF-3A | .500 | .190-24UNJS | .169 | .375 | 1.250 | — | MS17299-046 |
| .190-32UNJF-3A | .500 | .190-24UNJS | .169 | .375 | 1.375 | — | MS17299-056 |
| .190-32UNJF-3A | .500 | .190-24UNJS | .169 | .375 | 1.500 | — | MS17299-066 |
| .190-32UNJF-3A | .500 | .190-24UNJS | .169 | .375 | 1.625 | — | MS17299-076 |
| .250-28UNJF-3A | .438 | .250-20UNJS | .226 | .500 | 1.000 | — | MS17300-011 |
| .250-28UNJF-3A | .438 | .250-20UNJS | .226 | .500 | 1.125 | MS17302-016 | MS17300-021 |
| .250-28UNJF-3A | .438 | .250-20UNJS | .226 | .500 | 1.250 | MS17302-026 | MS17300-031 |
| .250-28UNJF-3A | .562 | .250-20UNJS | .226 | .500 | 1.375 | MS17302-036 | MS17300-051 |
| .250-28UNJF-3A | .562 | .250-20UNJS | .226 | .500 | 1.500 | MS17302-046 | — |
| .250-28UNJF-3A | .688 | .250-20UNJS | .226 | .500 | 1.500 | — | MS17300-061 |
| .250-28UNJF-3A | .562 | .250-20UNJS | .226 | .500 | 1.625 | MS17302-056 | MS17300-071 |
| .250-28UNJF-3A | .562 | .250-20UNJS | .226 | .500 | 1.750 | MS17302-066 | MS17300-081 |
| .250-28UNJF-3A | .562 | .250-20UNJS | .226 | .500 | 1.875 | MS17302-076 | MS17300-091 |
| .250-28UNJF-3A | .562 | .250-20UNJS | .226 | .500 | 2.000 | MS17302-086 | MS17300-101 |
| .250-28UNJF-3A | .562 | .250-20UNJS | .226 | .500 | 2.125 | MS17302-096 | MS17300-111 |
| .250-28UNJF-3A | .562 | .250-20UNJS | .226 | .500 | 2.250 | MS17302-106 | MS17300-121 |
| .250-28UNJF-3A | .562 | .250-20UNJS | .226 | .500 | 2.375 | MS17302-111 | MS17300-126 |
| .250-28UNJF-3A | .562 | .250-20UNJS | .226 | .500 | 2.500 | MS17302-116 | MS17300-131 |
| .250-28UNJF-3A | .562 | .250-20UNJS | .226 | .500 | 2.750 | MS17302-126 | MS17300-141 |
| .250-28UNJF-3A | .562 | .250-20UNJS | .226 | .500 | 3.000 | MS17302-136 | MS17300-151 |
| .250-28UNF-3A | .562 | .250-20UNS | .226 | .688 | 3.250 | AN128538 | — |
| .250-28UNF-3A | .562 | .250-20UNS | .226 | .688 | 3.500 | AN128550 | — |
| .250-28UNF-3A | .562 | .250-20UNS | .226 | .688 | 3.750 | AN128562 | — |
| .250-28UNF-3A | .562 | .250-20UNS | .226 | .688 | 4.000 | AN128574 | — |
| .250-28UNF-3A | .562 | .250-20UNS | .226 | .688 | 4.250 | AN128586 | — |
| .250-28UNF-3A | .562 | .250-20UNS | .226 | .688 | 4.500 | AN128598 | — |
| .250-28UNF-3A | .562 | .250-20UNS | .226 | .688 | 4.750 | AN128610 | — |
| .250-28UNF-3A | .562 | .250-20UNS | .226 | .688 | 5.000 | AN128622 | — |



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TABLE I PART NUMBERS (CONTINUED)

| A NUT END THREAD SIZE | B | C STUD END THREAD SIZE 2/ | D (NOM) | E | LENGTH (NOM) | 1/ PART NUMBER | |
|---|---|---|---|---|---|---|---|
| | | | | | | DRILLED | UNDRILLED |
| .3125-24UNJF-3A | .500 | .3125-18UNJS | .284 | .625 | 1.250 | MS17303-011 | MS17301-011 |
| .3125-24UNJF-3A | .562 | .3125-18UNJS | .284 | .625 | 1.375 | MS17303-021 | MS17301-021 |
| .3125-24UNJF-3A | .500 | .3125-18UNJS | .284 | .625 | 1.500 | MS17303-031 | MS17301-031 |
| .3125-24UNJF-3A | .625 | .3125-18UNJS | .284 | .625 | 1.625 | MS17303-041 | MS17301-051 |
| .3125-24UNJF-3A | .625 | .3125-18UNJS | .284 | .625 | 1.750 | MS17303-051 | MS17301-066 |
| .3125-24UNJF-3A | .625 | .3125-18UNJS | .284 | .625 | 1.875 | MS17303-061 | MS17301-081 |
| .3125-24UNJF-3A | .625 | .3125-18UNJS | .284 | .625 | 2.000 | MS17303-071 | MS17301-091 |
| .3125-24UNJF-3A | .625 | .3125-18UNJS | .284 | .625 | 2.125 | MS17303-081 | MS17301-101 |
| .3125-24UNJF-3A | .625 | .3125-18UNJS | .284 | .625 | 2.250 | MS17303-091 | MS17301-111 |
| .3125-24UNJF-3A | .625 | .3125-18UNJS | .284 | .625 | 2.375 | MS17303-101 | MS17301-121 |
| .3125-24UNJF-3A | .625 | .3125-18UNJS | .284 | .625 | 2.500 | MS17303-106 | MS17301-126 |
| .3125-24UNJF-3A | .625 | .3125-18UNJS | .284 | .625 | 2.750 | MS17303-116 | MS17301-136 |
| .3125-24UNJF-3A | .625 | .3125-18UNJS | .284 | .625 | 3.000 | MS17303-126 | MS17301-146 |
| .3125-24UNJF-3A | .625 | .3125-18UNJS | .284 | .625 | 3.250 | MS17303-136 | MS17301-156 |
| .3125-24UNJF-3A | .625 | .3125-18UNJS | .284 | .625 | 3.500 | MS17303-146 | MS17301-166 |
| .3125-24UNJF-3A | .625 | .3125-18UNJS | .284 | .625 | 3.750 | MS17303-156 | MS17301-176 |
| .3125-24UNJF-3A | .625 | .3125-18UNJS | .284 | .625 | 4.000 | MS17303-166 | MS17301-186 |
| .3125-24UNF-3A | .625 | .3125-18UNS | .284 | .844 | 4.250 | AN128898 | — |
| .3125-24UNF-3A | .625 | .3125-18UNS | .284 | .844 | 4.500 | AN128910 | — |
| .3125-24UNF-3A | .625 | .3125-18UNS | .284 | .844 | 4.750 | AN128922 | — |
| .3125-24UNF-3A | .625 | .3125-18UNS | .284 | .844 | 5.000 | AN128934 | — |
| .375-24UNF-3A | .562 | .375-16UNS | .347 | .750 | 1.438 | — | AN126600 |
| .375-24UNF-3A | .562 | .375-16UNS | .347 | .812 | 1.500 | — | AN126606 |
| .375-24UNF-3A | .562 | .375-16UNS | .347 | .938 | 1.625 | — | AN126612 |
| .375-24UNF-3A | .938 | .375-16UNS | .347 | .750 | 1.781 | — | AN126624 |
| .375-24UNF-3A | .938 | .375-16UNS | .347 | .750 | 1.875 | — | AN126630 |
| .375-24UNF-3A | .938 | .375-16UNS | .347 | .938 | 2.062 | — | AN126642 |

1/ REFER TO STANDARD SIZE COLUMN ON APPLICABLE DOCUMENT FOR PITCH DIAMETER OF PREFERRED PARTS LISTED.

2/ IT IS PERMISSIBLE TO USE UNDERSIZE AND OVERSIZE THREAD PITCH DIAMETERS SPECIFIED ON THE APPLICABLE DOCUMENTS IN PLACE OF STANDARD PITCH DIAMETERS IF REQUIRED TO ACHIEVE NECESSARY TORQUE VALUES DURING INITIAL INSTALLATION. HOWEVER, THIS PRACTICE IS NOT RECOMMENDED.

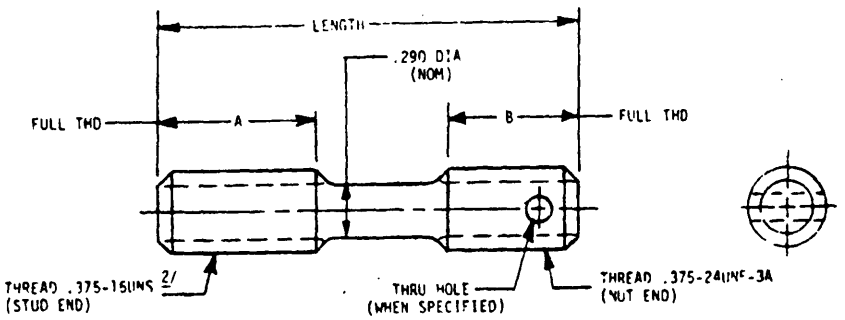


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SECTION 303

STUD, STRAIGHT, NECKED

APPLICABLE DOCUMENTS: AN126881 THRU AN127192, AN129293 THRU AN129604




| MATERIAL | HARDNESS | PROTECTIVE FINISH |
|---|---|---|
| STEEL, ALLOY | ROCKWELL C-26-32 | CADMIUM PLATE |

TABLE I. PART NUMBERS

| A | B | LENGTH (NOM) | PART NUMBER 1/ | |
|---|---|---|---|---|
| | | | DRILLED | UNDRILLED |
| .750 | .625 | 1.625 | AN129312 | AN126900 |
| .938 | .562 | 1.750 | AN129324 | AN126912 |
| ↑ | .688 | 1.875 | AN129336 | AN126924 |
| ↑ | ↑ | 1.938 | ——— | AN126930 |
| ↑ | ↑ | 2.000 | AN129348 | AN126936 |
| ↑ | ↑ | 2.062 | ——— | AN126942 |
| ↑ | ↑ | 2.125 | AN129360 | AN126948 |
| ↑ | ↑ | 2.250 | AN129372 | AN126960 |
| ↑ | ↑ | 2.312 | ——— | AN126966 |
| ↑ | ↑ | 2.375 | AN129384 | AN126972 |
| ↑ | ↑ | 2.500 | AN129396 | AN126984 |
| ↑ | ↑ | 2.625 | ——— | AN126990 |
| ↑ | ↑ | 2.750 | AN129408 | AN126996 |
| ↑ | ↑ | 2.875 | ——— | AN127002 |
| ↑ | ↑ | 3.000 | AN129420 | AN127008 |
| ↑ | ↑ | 3.250 | AN129432 | AN127020 |
| ↑ | ↑ | 3.500 | AN129444 | AN127032 |
| ↓ | ↑ | 3.750 | AN129456 | AN127044 |
| .938 | ↑ | 3.875 | ——— | AN127050 |
| 1.375 | ↑ | 4.000 | AN129468 | AN127056 |
| ↑ | ↑ | 4.250 | AN129480 | AN127068 |
| ↑ | ↑ | 4.500 | AN129492 | AN127080 |
| ↑ | ↑ | 4.750 | AN129504 | AN127092 |
| ↑ | ↑ | 5.000 | AN129516 | AN127104 |
| ↓ | ↓ | 5.250 | AN129528 | AN127116 |
| 1.375 | .688 | 5.500 | AN129540 | AN127128 |

1/ REFER TO STANDARD SIZE COLUMN ON APPLICABLE DOCUMENT FOR PITCH DIAMETER OF PREFERRED PARTS LISTED.

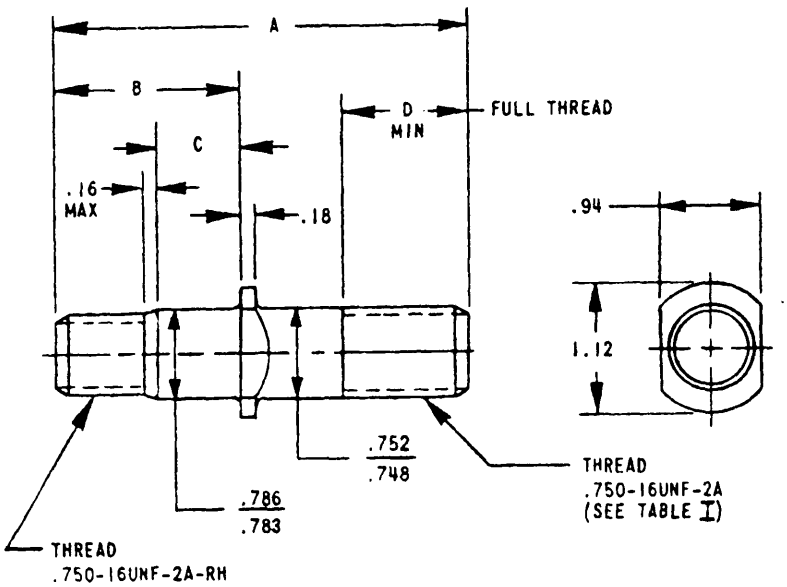
2/ IT IS PERMISSIBLE TO USE UNDERSIZE AND OVERSIZE THREAD PITCH DIAMETERS SPECIFIED ON THE APPLICABLE DOCUMENTS IN PLACE OF STANDARD PITCH DIAMETERS IF REQUIRED TO ACHIEVE NECESSARY TORQUE VALUES DURING INITIAL INSTALLATION. HOWEVER, THIS PRACTICE IS NOT RECOMMENDED



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SECTION 401

STUD, SHOULDERED, WHEEL MOUNTING
APPLICABLE DOCUMENT: MS51950




| MATERIAL | TENSILE STRENGTH (PSI) MIN | PROTECTIVE FINISH |
|---|---|---|
| STEEL, ALLOY | 120,000 | ZINC OR CADMIUM PLATE |

TABLE I

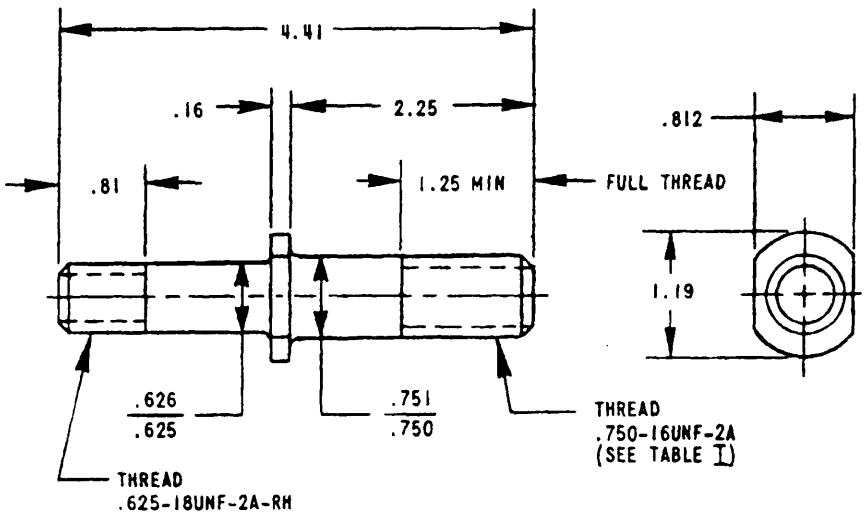
| THREAD | A | B | C | D | MS51950 |
|---|---|---|---|---|---|
| LH | 3.59 | 2.06 | .88 | 1.14 | -1 |
| RH | 3.59 | 2.06 | .88 | 1.14 | -2 |
| LH | 3.81 | 1.63 | .72 | 1.20 | -3 |
| RH | 3.81 | 1.63 | .72 | 1.20 | -4 |
| LH | 4.03 | 2.50 | 1.50 | 1.14 | -5 |
| RH | 4.03 | 2.50 | 1.50 | 1.14 | -6 |



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SECTION 501

STUD, SHOULDERED AND STEPPED, WHEEL MOUNTING
(.625-18UNF-2A X.750-16UNF-2A)
APPLICABLE DOCUMENT: MS52122




| MATERIAL | TENSILE STRENGTH (PSI) MIN | PROTECTIVE FINISH |
|---|---|---|
| STEEL, ALLOY | 120,000 | ZINC OR CADMIUM PLATE |

TABLE I

| THREAD | MS52122 |
|---|---|
| LH | -1 |
| RH | -2 |

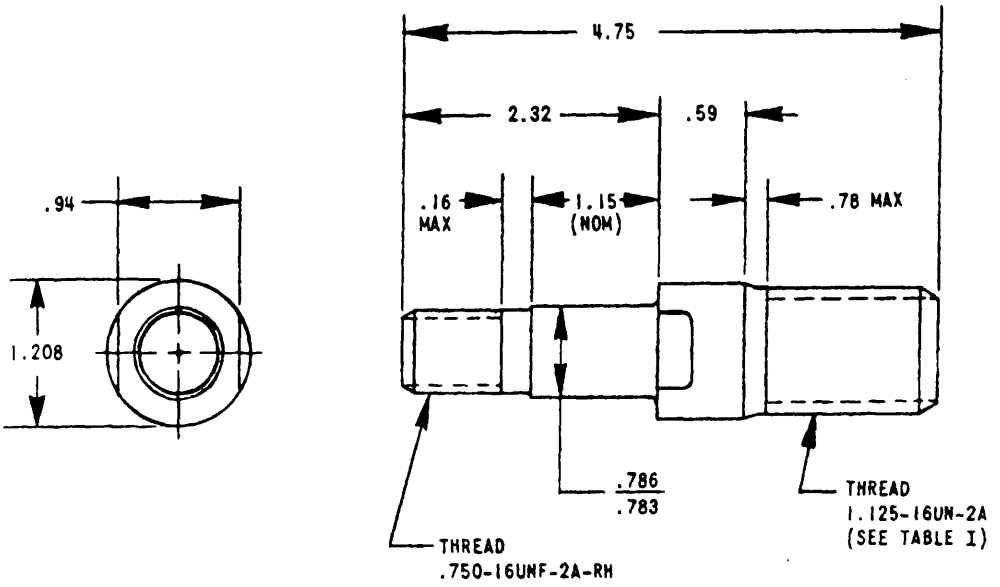


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SECTION 502

STUD, SHOULDERED AND STEPPED, WHEEL MOUNTING
(.750-16UNF-2A X 1.125-16UN-2A)

APPLICABLE DOCUMENT: MS51951




| MATERIAL | TENSILE STRENGTH (PSI) MIN | PROTECTIVE FINISH |
|---|---|---|
| STEEL, ALLOY | 120.000 | ZINC OR CADMIUM PLATE |

TABLE I

| THREAD | MS51951 |
|---|---|
| LH | -1 |
| RH | -2 |

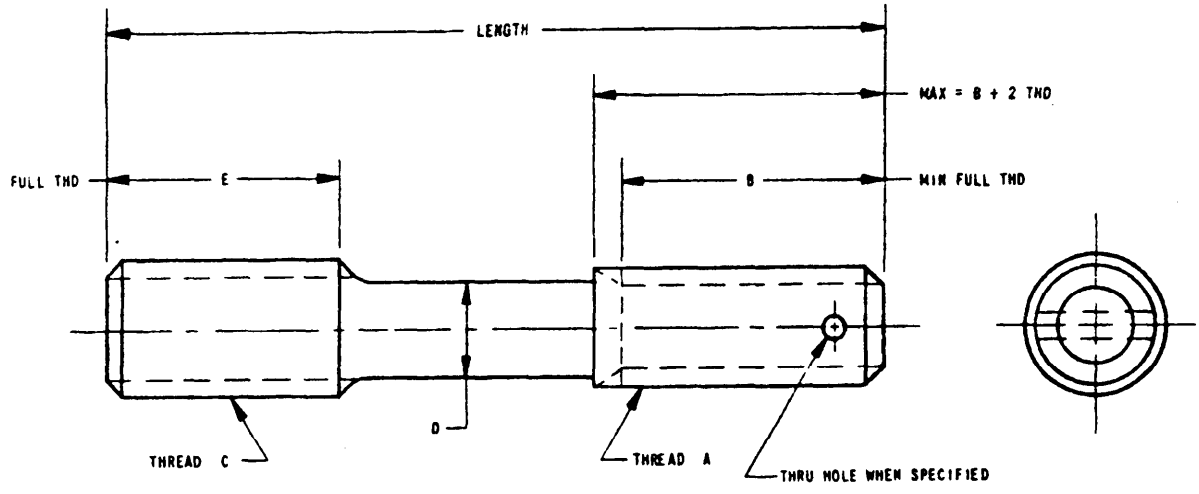


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SECTION 601

STUD, STEPPED, 1.5 DIA ENGAGEMENT

APPLICABLE DOCUMENTS: AN152601 THRU AN152900,
MS17293, MS17294, MS 17295 MS17296, MS17297, MS17298




| MATERIAL | HARDNESS | PROTECTIVE FINISH |
|---|---|---|
| STEEL, ALLOY | ROCKWELL C-26-32 | CADMIUM PLATE |



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TABLE I PART NUMBERS

| A NUT END THREAD SIZE | B | C STUD END THREAD SIZE 2/ | D (NOM) | E (NOM) | LENGTH | 1/ PART NUMBER | |
|---|---|---|---|---|---|---|---|
| | | | | | | DRILLED | UNDRILLED |
| .190-32UNF-3A | .375 | .250-20UNS | .169 | .375 | .875 | AN152620 | — |
| .190-32UNF-3A | .500 | .250-20UNS | .169 | .375 | 1.000 | AN152632 | — |
| .190-32UNF-3A | .500 | .250-20UNS | .169 | .375 | 1.125 | AN152644 | — |
| .190-32UNF-3A | .500 | .250-20UNS | .169 | .375 | 1.250 | AN152656 | — |
| .190-32UNF-3A | .500 | .250-20UNS | .169 | .375 | 1.375 | AN152668 | — |
| .190-32UNF-3A | .500 | .250-20UNS | .169 | .375 | 1.500 | AN152680 | — |
| .190-32UNF-3A | .500 | .250-20UNS | .169 | .375 | 1.625 | AN152692 | — |
| .190-32UNJF-3A | .375 | .250-20UNJS | .169 | .375 | .875 | — | MS17293-011 |
| .190-32UNJF-3A | .500 | .250-20UNJS | .169 | .375 | 1.000 | — | MS17293-021 |
| .190-32UNJF-3A | .500 | .250-20UNJS | .169 | .375 | 1.125 | — | MS17293-031 |
| .190-32UNJF-3A | .500 | .250-20UNJS | .169 | .375 | 1.250 | — | MS17293-041 |
| .190-32UNJF-3A | .500 | .250-20UNJS | .169 | .375 | 1.375 | — | MS17293-051 |
| .190-32UNJF-3A | .500 | .250-20UNJS | .169 | .375 | 1.500 | — | MS17293-061 |
| .190-32UNJF-3A | .500 | .250-20UNJS | .169 | .375 | 1.625 | — | MS17293-071 |
| .250-28UNJF-3A | .438 | .3125-18UNJS | .226 | .469 | 1.031 | MS17297-011 | MS17294-011 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .226 | .469 | 1.156 | MS17297-021 | MS17294-021 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .226 | .469 | 1.281 | MS17297-031 | — |
| .250-28UNJF-3A | .656 | .3125-18UNJS | .226 | .469 | 1.281 | — | MS17294-031 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .226 | .469 | 1.406 | MS17297-041 | — |
| .250-28UNJF-3A | .688 | .3125-18UNJS | .226 | .469 | 1.406 | — | MS17294-041 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .226 | .469 | 1.531 | MS17297-051 | MS17294-051 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .226 | .469 | 1.656 | MS17297-061 | MS17294-061 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .226 | .469 | 1.781 | MS17297-071 | MS17294-071 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .226 | .469 | 1.906 | MS17297-081 | MS17294-081 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .226 | .469 | 2.031 | MS17297-091 | MS17294-091 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .226 | .469 | 2.156 | MS17297-101 | MS17294-101 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .226 | .469 | 2.344 | MS17297-111 | MS17294-111 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .226 | .469 | 2.594 | MS17297-121 | MS17294-121 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .226 | .469 | 2.844 | MS17297-131 | MS17294-131 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .226 | .469 | 3.094 | MS17297-141 | MS17294-141 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .226 | .469 | 3.344 | MS17297-151 | MS17294-151 |



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TABLE I PART NUMBERS (CONTINUED)

| A NUT END THREAD SIZE | B | C STUD END THREAD SIZE 2/ | D (NOM) | E (NOM) | LENGTH | 1/ PART NUMBER | |
|---|---|---|---|---|---|---|---|
| | | | | | | DRILLED | UNDRILLED |
| .3125-24UNJF-3A | .562 | .375-16UNJS | .284 | .562 | 1.250 | MS17298-016 | MS17295-016 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .284 | .562 | 1.375 | MS17298-026 | MS17295-026 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .284 | .562 | 1.500 | MS17298-036 | — |
| .3125-24UNJF-3A | .719 | .375-16UNJS | .284 | .562 | 1.500 | — | MS17295-036 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .284 | .562 | 1.625 | MS17298-046 | — |
| .3125-24UNJF-3A | .812 | .375-16UNJS | .284 | .562 | 1.625 | — | MS17295-051 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .284 | .562 | 1.750 | MS17298-056 | MS17295-061 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .284 | .562 | 1.875 | MS17298-066 | MS17295-071 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .284 | .562 | 2.000 | MS17298-076 | MS17295-081 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .284 | .562 | 2.125 | MS17298-086 | MS17295-091 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .284 | .562 | 2.250 | MS17298-096 | MS17295-101 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .284 | .562 | 2.562 | MS17298-111 | MS17295-116 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .284 | .562 | 2.812 | MS17298-121 | MS17295-126 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .284 | .562 | 3.062 | MS17298-131 | MS17295-136 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .284 | .562 | 3.312 | MS17298-141 | MS17295-146 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .284 | .562 | 3.562 | MS17298-151 | MS17295-156 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .284 | .562 | 3.812 | MS17298-161 | MS17295-166 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .284 | .562 | 4.062 | MS17298-171 | MS17295-176 |
| .375-24UNJF-3A | .625 | .4375-14UNJS | .347 | .656 | 1.406 | — | MS17296-016 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .347 | .656 | 1.531 | — | MS17296-026 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .347 | .656 | 1.656 | — | MS17296-036 |
| .375-24UNJF-3A | .938 | .4375-14UNJS | .347 | .656 | 1.750 | — | MS17296-046 |
| .375-24UNJF-3A | .938 | .4375-14UNJS | .347 | .656 | 1.844 | — | MS17296-056 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .347 | .656 | 2.031 | — | MS17296-066 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .347 | .656 | 2.281 | — | MS17296-086 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .347 | .656 | 2.531 | — | MS17296-101 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .347 | .656 | 2.781 | — | MS17296-111 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .347 | .656 | 3.031 | — | MS17296-121 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .347 | .656 | 3.281 | — | MS17296-131 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .347 | .656 | 3.531 | — | MS17296-141 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .347 | .656 | 3.781 | — | MS17296-151 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .347 | .656 | 4.031 | — | MS17296-161 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .347 | .656 | 4.281 | — | MS17296-171 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .347 | .656 | 4.531 | — | MS17296-181 |

1/ REFER TO STANDARD SIZE COLUMN ON APPLICABLE DOCUMENT FOR PITCH DIAMETER OF PREFERRED PARTS LISTED.

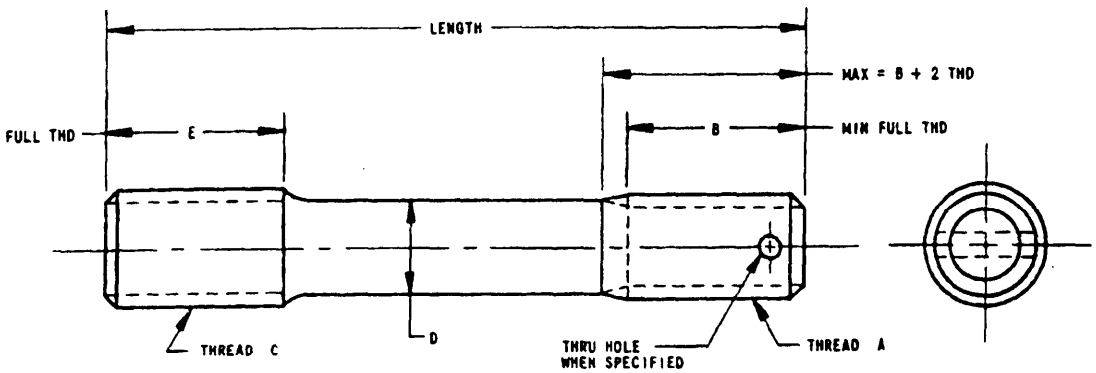
2/ IT IS PERMISSIBLE TO USE UNDERSIZE AND OVERSIZE THREAD PITCH DIAMETERS SPECIFIED ON THE APPLICABLE DOCUMENTS IN PLACE OF STANDARD PITCH DIAMETERS IF REQUIRED TO ACHIEVE NECESSARY TORQUE VALUES DURING INITIAL INSTALLATION. HOWEVER, THIS PRACTICE IS NOT RECOMMENDED



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SECTION 602

STUD, STEPPED, 2 DIA ENGAGEMENT
APPLICABLE DOCUMENTS: MS9827 THRU MS9840




| MATERIAL | HARDNESS | PROTECTIVE FINISH |
|---|---|---|
| STEEL, ALLOY | ROCKWELL C-26-32 | CADMIUM PLATE |

TABLE I PART NUMBERS

| A NUT END THREAD SIZE | B | C 2/ STUD END THREAD SIZE | D (NOM) | E | LENGTH (NOM) | 1/ PART NUMBER | |
|---|---|---|---|---|---|---|---|
| | | | | | | DRILLED | UNDRILLED |
| .190-32UNJF-3A | .375 | .250-20UNJS | .169 | .500 | 1.000 | MS9834-011 | MS9827-011 |
| .190-32UNJF-3A | .500 | .250-20UNJS | .169 | .500 | 1.125 | MS9834-021 | MS9827-021 |
| .190-32UNJF-3A | .500 | .250-20UNJS | .169 | .500 | 1.250 | MS9834-031 | MS9827-031 |
| .190-32UNJF-3A | .500 | .250-20UNJS | .169 | .500 | 1.375 | MS9834-041 | MS9827-041 |
| .190-32UNJF-3A | .500 | .250-20UNJS | .169 | .500 | 1.500 | MS9834-051 | MS9827-051 |
| .190-32UNJF-3A | .500 | .250-20UNJS | .169 | .500 | 1.625 | MS9834-061 | MS9827-061 |
| .190-32UNJF-3A | .500 | .250-20UNJS | .169 | .500 | 1.750 | MS9834-071 | MS9827-071 |
| .250-28UNJF-3A | .500 | .3125-18UNJS | .227 | .625 | 1.250 | MS9835-016 | MS9828-016 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 1.375 | MS9835-026 | MS9828-026 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 1.500 | MS9835-036 | MS9828-036 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 1.625 | MS9835-046 | MS9828-046 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 1.750 | MS9835-056 | MS9828-056 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 1.875 | MS9835-066 | MS9828-066 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 2.000 | MS9835-076 | MS9828-076 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 2.125 | MS9835-086 | MS9828-086 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 2.250 | MS9835-096 | MS9828-096 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 2.375 | MS9835-106 | MS9828-106 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 2.500 | MS9835-116 | MS9828-116 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 2.625 | MS9835-126 | MS9828-126 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 2.750 | MS9835-136 | MS9828-136 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 2.875 | MS9835-146 | MS9828-146 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 3.000 | MS9835-156 | MS9828-156 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 3.250 | MS9835-166 | MS9828-166 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 3.375 | MS9835-171 | — |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 3.500 | MS9835-176 | MS9828-176 |
| .250-28UNJF-3A | .562 | .3125-18UNJS | .227 | .625 | 3.625 | MS9835-181 | — |
| .3125-24UNJF-3A | .500 | .375-16UNJS | .285 | .750 | 1.375 | MS9836-011 | MS9829-011 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .285 | .750 | 1.500 | MS9836-021 | MS9829-021 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .285 | .750 | 1.625 | MS9836-031 | MS9829-031 |



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TABLE I PART NUMBERS (CONTINUED)

| A NUT END THREAD SIZE | B | C STUD END 2/ THREAD SIZE | D (NOM) | E | LENGTH (NOM) | 1/ PART NUMBER | |
|---|---|---|---|---|---|---|---|
| | | | | | | DRILLED | UNDRILLED |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .285 | .750 | 1.750 | MS9836-041 | MS9829-046 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .285 | .750 | 1.875 | MS9836-051 | MS9829-056 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .285 | .750 | 2.000 | MS9836-061 | MS9829-066 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .285 | .750 | 2.125 | MS9836-071 | MS9829-076 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .285 | .750 | 2.250 | MS9836-081 | MS9829-086 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .285 | .750 | 2.375 | MS9836-091 | MS9829-096 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .285 | .750 | 2.500 | MS9836-101 | MS9829-106 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .285 | .750 | 2.750 | MS9836-111 | MS9829-116 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .285 | .750 | 3.000 | MS9836-121 | MS9829-126 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .285 | .750 | 3.250 | MS9836-131 | MS9829-136 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .285 | .750 | 3.500 | MS9836-141 | MS9829-146 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .285 | .750 | 3.750 | MS9836-151 | MS9829-156 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .285 | .750 | 4.000 | MS9836-161 | MS9829-166 |
| .3125-24UNJF-3A | .625 | .375-16UNJS | .285 | .750 | 4.250 | MS9836-171 | MS9829-176 |
| .375-24UNJF-3A | .625 | .4375-14UNJS | .348 | .875 | 1.625 | MS9837-016 | MS9830-016 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 1.750 | MS9837-026 | MS9830-026 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 1.875 | MS9837-036 | MS9830-036 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 2.000 | MS9837-046 | MS9830-051 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 2.125 | MS9837-056 | MS9830-061 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 2.250 | MS9837-066 | MS9830-071 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 2.375 | MS9837-076 | MS9830-081 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 2.500 | MS9837-086 | MS9830-091 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 2.750 | MS9837-101 | MS9830-106 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 3.000 | MS9837-111 | MS9830-116 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 3.250 | MS9837-121 | MS9830-126 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 3.500 | MS9837-131 | MS9830-136 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 3.750 | MS9837-141 | MS9830-146 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 4.000 | MS9837-151 | MS9830-156 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 4.250 | MS9837-161 | MS9830-166 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 4.500 | MS9837-171 | MS9830-176 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 4.750 | MS9837-181 | MS9830-186 |
| .375-24UNJF-3A | .688 | .4375-14UNJS | .348 | .875 | 5.000 | MS9837-191 | MS9830-196 |
| .4375-20UNJF-3A | .625 | .500-13UNJS | .405 | 1.000 | 1.750 | MS9838-011 | MS9831-011 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 1.875 | MS9838-021 | MS9831-021 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 2.000 | MS9838-031 | MS9831-031 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 2.125 | MS9838-041 | MS9831-041 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 2.250 | MS9838-051 | MS9831-051 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 2.375 | MS9838-061 | MS9831-061 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 2.500 | MS9838-071 | MS9831-071 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 2.625 | MS9838-081 | MS9831-081 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 2.750 | MS9838-091 | MS9831-091 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 3.000 | MS9838-101 | MS9831-101 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 3.250 | MS9838-111 | MS9831-111 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 3.500 | MS9838-121 | MS9831-121 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 3.750 | MS9838-131 | MS9831-131 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 4.000 | MS9838-141 | MS9831-141 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 4.250 | MS9838-151 | MS9831-151 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 4.500 | MS9838-161 | MS9831-161 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 4.750 | MS9838-171 | MS9831-171 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 5.000 | MS9838-181 | MS9831-181 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 5.250 | MS9838-191 | MS9831-191 |
| .4375-20UNJF-3A | .750 | .500-13UNJS | .405 | 1.000 | 5.500 | MS9838-201 | MS9831-201 |



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TABLE I PART NUMBERS (CONTINUED)

| A NUT END THREAD SIZE | B | C 2/ STUD END THREAD SIZE | D (NOM) | E | LENGTH (NOM) | 1/ PART NUMBER | |
|---|---|---|---|---|---|---|---|
| | | | | | | DRILLED | UNDRILLED |
| .500-20UNJF-3A | .750 | .5625-12UNJS | .467 | 1.125 | 2.000 | MS9839-011 | MS9832-011 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 2.125 | MS9839-021 | MS9832-021 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 2.250 | MS9839-031 | MS9832-031 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 2.375 | MS9839-041 | MS9832-041 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 2.500 | MS9839-051 | MS9832-051 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 2.625 | MS9839-061 | MS9832-061 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 2.750 | MS9839-071 | MS9832-071 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 2.875 | — | MS9832-081 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 3.000 | MS9839-086 | MS9832-086 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 3.250 | MS9839-096 | MS9832-096 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 3.500 | MS9839-106 | MS9832-106 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 3.750 | MS9839-116 | MS9832-116 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 4.000 | MS9839-126 | MS9832-126 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 4.250 | MS9839-136 | MS9832-136 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 4.500 | MS9839-146 | MS9832-146 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 4.750 | MS9839-156 | MS9832-156 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 5.000 | MS9839-166 | MS9832-166 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 5.250 | MS9839-176 | MS9832-176 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 5.500 | MS9839-186 | MS9832-186 |
| .500-20UNJF-3A | .875 | .5625-12UNJS | .467 | 1.125 | 5.750 | MS9839-196 | MS9832-196 |
| .5625-18UNJF-3A | .875 | .625-11UNJS | .526 | 1.250 | 2.250 | MS9840-016 | MS9833-016 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 2.375 | MS9840-026 | MS9833-026 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 2.500 | MS9840-036 | MS9833-036 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 2.625 | MS9840-046 | MS9833-046 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 2.750 | MS9840-056 | MS9833-056 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 2.875 | MS9840-066 | MS9833-066 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 3.000 | MS9840-076 | MS9833-076 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 3.250 | MS9840-086 | MS9833-086 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 3.500 | MS9840-096 | MS9833-096 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 3.750 | MS9840-106 | MS9833-106 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 4.000 | MS9840-116 | MS9833-116 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 4.250 | MS9840-126 | MS9833-126 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 4.500 | MS9840-136 | MS9833-136 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 4.750 | MS9840-146 | MS9833-146 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 5.000 | MS9840-156 | MS9833-156 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 5.250 | MS9840-166 | MS9833-166 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 5.500 | MS9840-176 | MS9833-176 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 5.750 | MS9840-186 | MS9833-186 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 6.000 | MS9840-196 | MS9833-196 |
| .5625-18UNJF-3A | 1.000 | .625-11UNJS | .526 | 1.250 | 6.250 | MS9840-206 | MS9833-206 |

1/ REFER TO STANDARD SIZE COLUMN ON APPLICABLE DOCUMENT FOR PITCH DIAMETER OF PREFERRED PARTS LISTED.

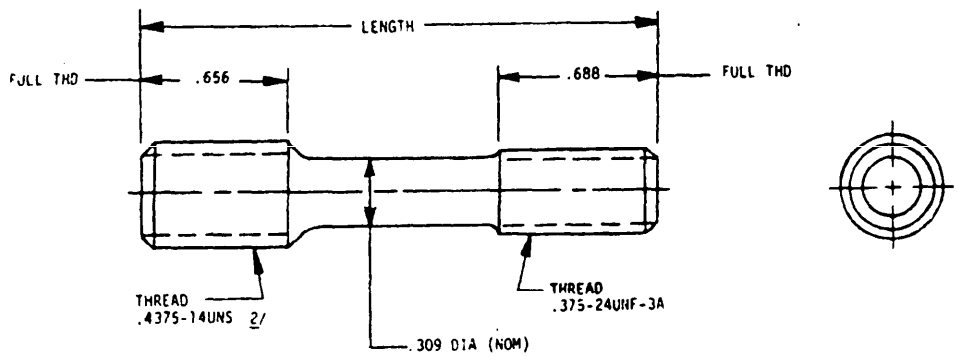
2/ IT IS PERMISSIBLE TO USE OVERSIZE THREAD PITCH DIAMETERS SPECIFIED ON THE APPLICABLE DOCUMENTS IN PLACE OF STANDARD PITCH DIAMETERS IF REQUIRED TO ACHIEVE NECESSARY TORQUE VALUES DURING INITIAL INSTALLATION. HOWEVER, THIS PRACTICE IS NOT RECOMMENDED.



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SECTION 603

STUD, STEPPED, NECKED, 1.5 DIA ENGAGEMENT .4375-14UNS X .375-24UNF-3A
APPLICABLE DOCUMENT: AN158901 THRU AN159200




| MATERIAL | HARDNESS | PROTECTIVE FINISH |
|---|---|---|
| STEEL, ALLOY | ROCKWELL C-26-32 | CADMIUM PLATE |

TABLE I PART NUMBERS

| LENGTH | PART NUMBER 1/ |
|---|---|
| 1.594 | AN158914 |
| 1.656 | AN158920 |
| 1.781 | AN158932 |
| 1.906 | AN158944 |
| 2.031 | AN158956 |
| 2.156 | AN158968 |
| 2.281 | AN158980 |
| 2.531 | AN158998 |
| 2.781 | AN159010 |
| 3.031 | AN159022 |
| 3.281 | AN159034 |
| 3.531 | AN159046 |
| 3.781 | AN159058 |
| 4.031 | AN159070 |
| 4.281 | AN159082 |
| 4.531 | AN159094 |

1/ REFER TO STANDARD SIZE COLUMN ON APPLICABLE DOCUMENT FOR PITCH DIAMETER OF PREFERRED PARTS LISTED.

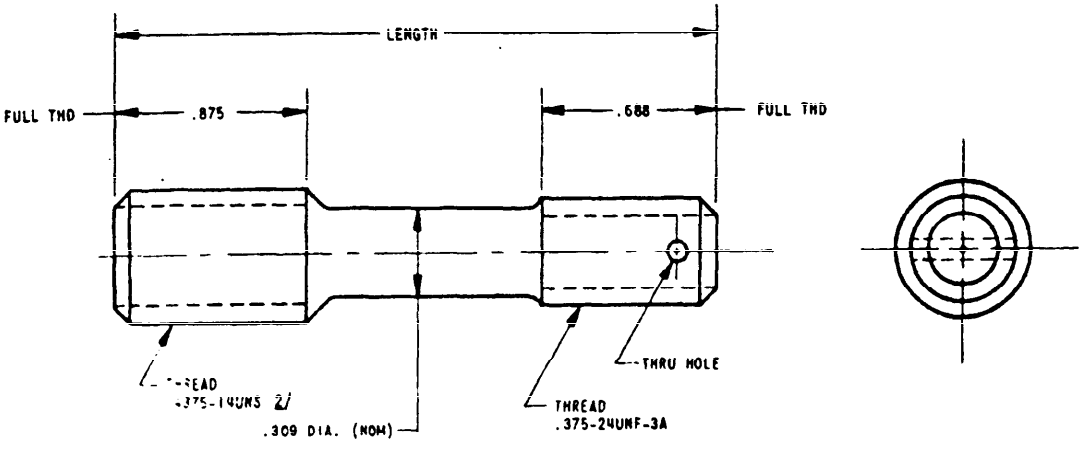
2/ IT IS PERMISSIBLE TO USE UNDERSIZE AND OVERSIZE THREAD PITCH DIAMETERS SPECIFIED ON THE APPLICABLE DOCUMENTS IN PLACE OF STANDARD PITCH DIAMETERS IF REQUIRED TO ACHIEVE NECESSARY TORQUE VALUES DURING INITIAL INSTALLATION. HOWEVER, THIS PRACTICE IS NOT RECOMMENDED.



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SECTION 604

STUD, STEPPED, DRILLED, NECKED, 2 DIA ENGAGEMENT .4375-14UNS X .375-24UNF-3A
APPLICABLE DOCUMENT: AN162501 THRU AN162800




| MATERIAL | HARDNESS | PROTECTIVE FINISH |
|---|---|---|
| STEEL, ALLOY | ROCKWELL C-26-32 | CADMIUM PLATE |

TABLE I PART NUMBERS

| LENGTH | PART NUMBER 1/ |
|---|---|
| 1.875 | AN162520 |
| 2.000 | AN162532 |
| 2.125 | AN162544 |
| 2.250 | AN162556 |
| 2.375 | AN162568 |
| 2.500 | AN162580 |
| 2.750 | AN162598 |
| 3.000 | AN162610 |
| 3.250 | AN162622 |
| 3.500 | AN162634 |
| 3.750 | AN162646 |
| 4.000 | AN162658 |
| 4.250 | AN162670 |
| 4.500 | AN162682 |
| 4.750 | AN162694 |

1/ REFER TO STANDARD SIZE COLUMN ON APPLICABLE DOCUMENT FOR PITCH DIAMETER OF PREFERRED PARTS LISTED.

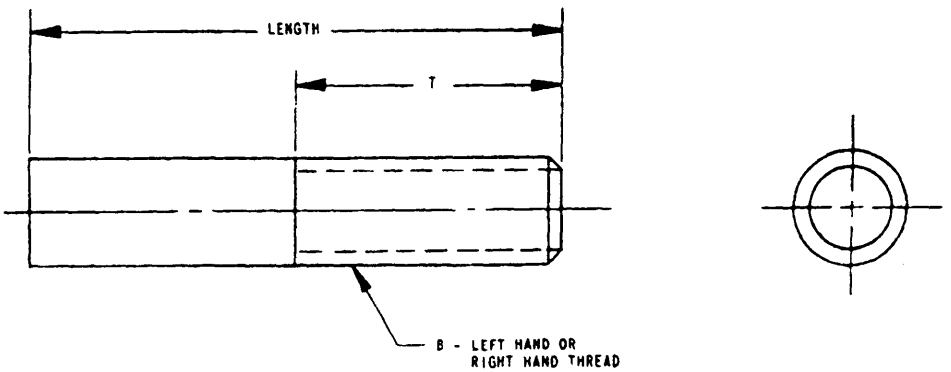
2/ IT IS PERMISSIBLE TO USE UNDERSIZE AND OVERSIZE THREAD PITCH DIAMETERS SPECIFIED ON THE APPLICABLE DOCUMENTS IN PLACE OF STANDARD PITCH DIAMETERS IF REQUIRED TO ACHIEVE NECESSARY TORQUE VALUES DURING FINAL INSTALLATION. HOWEVER, THIS PRACTICE IS NOT RECOMMENDED.



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SECTION 701

STUB (STUD) WELDING, TURNBUCKLE (GENERAL PURPOSE, MANUAL WELD APPLICATION)
APPLICABLE DOCUMENT: MS27951




| MATERIAL | PROTECTIVE FINISH |
|---|---|
| STEEL (FORGED) | ZINC COAT |

TABLE I

| B THREAD SIZE UNC-2A | LENGTH REF | T REF | TURNBUCKLE CLEAR OPENING REF | RECOMMENDED WORKING LOAD LBS. 1/ | MS27951 | |
|---|---|---|---|---|---|---|
| | | | | | LEFT HAND | RIGHT HAND |
| .250-20 | 4.75 | 2.50 | 4.00 | 500 | -1 | -2 |
| .3125-18 | 5.38 | 2.88 | 4.50 | 700 | -3 | -4 |
| .375-16 | 6.75 | 3.75 | 6.00 | 1,040 | -5 | -6 |
| .500-13 | 7.00<br>8.50<br>10.00 | 4.00<br>5.50<br>7.00 | 6.00<br>9.00<br>12.00 | 1,800 | -7<br>-9<br>-11 | -8<br>-10<br>-12 |
| .625-11 | 7.75<br>9.25<br>10.75<br>13.75 | 4.25<br>5.75<br>7.25<br>10.25 | 6.00<br>9.00<br>12.00<br>18.00 | 2,700 | -13<br>-15<br>-17<br>-19 | -14<br>-16<br>-18<br>-20 |
| .750-10 | 8.00<br>11.00<br>14.00 | 4.50<br>7.50<br>10.50 | 6.00<br>12.00<br>18.00 | 4,000 | -21<br>-23<br>-25 | -22<br>-24<br>-26 |
| 1.000-8 | 9.00<br>15.00<br>18.00 | 5.00<br>11.00<br>14.00 | 6.00<br>18.00<br>24.00 | 7,600 | -31<br>-33<br>-35 | -32<br>-34<br>-36 |
| 1 250-7 | 9.50<br>15.50<br>18.50 | 5.50<br>11.50<br>14.50 | 6.00<br>18.00<br>24.00 | 12,000 | -37<br>-39<br>-41 | -38<br>-40<br>-42 |
| 1 500-6 | 19.25 | 14.75 | 24.00 | 17,000 | -43 | -44 |
| 2 000-4-1/2 | 20.50 | 15.00 | 24.00 | 30,000 | -45 | -46 |
| 2.500-4 | 22.00 | 16.00 | 24.00 | 48,400 | -47 | -48 |

1/ - RECOMMENDED WORKING LOAD (SAFETY FACTOR 5 TO 1) BASED ON BREAKING STRENGTH AT THREAD ROOT.

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